

Enhancing Multilingual AI

Analyzing and Modeling Educational Content with the FineWeb-C Dataset

Subject Project



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By

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Abstract

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1 Introduction

1.1 Introduction to Multilingual AI and Related Work

Every minute, the digital world generates an unimaginable amount of information, from news articles to casual social media posts. How do we efficiently sift through the noise to find what is truly meaningful or useful? To address this, data labeling (or data annotation) is a crucial, yet costly process. Traditional methods, such as CAPTCHA, require humans to make decisions on specific pieces of data. As the demand for big data models continues to increase, so does the need for high-quality data and reliable data labels. In recent years, researchers have explored automated data labeling where AI models predict these labels. This approach, where LLMs themselves act as annotators, is a rapidly growing field of study [28], with many models proving effective at the task.

A essential moment for automated data labeling in Natural Language Processing (NLP) came in 2017, when Google researchers revolutionized the field with their paper "Attention is All You Need," introducing the transformer and the attention mechanism [5]. Following this, in 2018, transformer-based encoder models were popularized with models like BERT [10]. With their strong language understanding and excellent text classification abilities, these models advanced automated data annotation.

Crucially, this advancement was well timed, aligning with the principles of Data-Centric AI [8], a philosophy that emphasizes data quality is often more important for pretraining than quantity [11, 23, 22, 29]. As a result, the primary challenge has shifted from simple labeling to the more complex task of evaluating the data quality and value, a process that these powerful models are now able to do at scale. However, this rapid progress has been concentrated on English, mainly due to the massive amount of available data. For many other small languages, data scarcity is a major hurdle, slowing down AI development and creating a gap in capability. As a result, finding methods to annotate content in languages with less data often through techniques like cross-lingual transfer learning [1], is essential for making the power of modern AI accessible and effective on a global scale.

The process of fine-tuning large pretrained models like BERT for specific classification tasks is well established [27]. Recent research, such as at Université de Bourgogne [?], highlights how fine-tuned BERT models improve accuracy in multi-label classification. Another prime example is the FineWeb-Edu dataset [?], a large corpus of web-sourced text annotated for educational content, using a small BERT-like encoder model [? ?]. This efficient data and model pipeline inspired our approach.

Building upon these advancements, this project explores the extent to which pretrained transformer-based encoders, specifically multilingual models like XLM-RoBERTa [2], can be fine-tuned for scalable and accurate labeling of educational text data. Our focus is on evaluating the educational quality of content from the FineWeb-2 dataset [12], with the goal of contributing to more efficient and consistent data annotation pipelines for complex, subjective tasks in a low resource language context.

The main challenge addressed in this project is the effective and efficient annotation of Danish educational data extracted from the FineWeb-C dataset. While traditional human annotation has long been the standard, recent advancements in language models offer scalable and consistent alternatives. This study evaluates the capabilities of fine-tuned

multilingual encoders, specifically XLM-RoBERTa, in annotating Danish educational content. To evaluate these capabilities, it is first essential to understand the nature and variability of human annotations, in this inherently subjective task, which sets the practical benchmark for automated systems.

1.2 Research Question and Testable Hypothesis

This study is structured around three central research questions that are linked to our methodological approach:

1. **Inter-annotator agreement**

What level of agreement can be expected when annotating the educational value of texts, and what factors may influence consistency and reliability across annotators and categories?

2. **Efficacy of Cross-Lingual Fine-Tuning Strategies**

What are the relative strengths and limitations of zero-shot, few-shot, and full fine-tuning strategies when adapting XLM-R to classify Danish educational content, particularly concerning the trade-off between predictive performance and the amount of target-language data required?

3. **Model Alignment with Human annotations**

To what extent can a fine-tuned XLM-R model replicate human performance in annotating the educational value of Danish texts, considering the inherent ambiguity and level of disagreement observed among human annotators?

Following the problem setting and research questions outlined above, the paper establishes the following testable hypotheses that guide our experimental setup:

1. **Inter-Annotator Agreement:**

- (a) We hypothesize that human inter-annotator agreement for educational content will be moderate to low due to the inherent subjectivity of the task. This will be evaluated by analyzing Krippendorff's Alpha and Mean Squared Difference among human annotators on the FineWeb-C dataset.
- (b) We predict that the annotations provided by the authors will show higher agreement than the Danish annotations from FineWeb-C. This will be validated by comparing Krippendorff's Alpha and MSD scores between the author annotations and the FineWeb-C annotations, supported by statistical tests.
- (c) We hypothesize that all levels of educational value will exhibit similar levels of disagreement among annotators. This will be investigated using a pooled analysis of inter-annotator agreement across different educational value levels.

2. **Efficacy of Cross-Lingual Fine-Tuning Strategies:**

- (a) We hypothesize that XLM-R will achieve notable performance in a zero-shot English-to-Danish transfer by leveraging its pre-trained multilingual representations, establishing a strong baseline for low-resource settings. This will be evaluated by training the model exclusively on English data and testing it directly on the Danish FineWeb-C dataset using metrics such as MSE, Accuracy, and weighted F1-score.
- (b) We hypothesize that incorporating a limited number of Danish-labeled samples (few-shot fine-tuning) will lead to a significant improvement in XLM-R's

performance compared to the zero-shot baseline. This will be tested by warm-starting the English-trained model and fine-tuning it with small quantities (e.g., 250, 1000, 2500 samples) of Danish data, with performance metrics compared against the zero-shot results.

- (c) We hypothesize that full Danish fine-tuning, utilizing the complete target-language dataset, will yield the strongest performance on Danish educational content. This will be established by training the model exclusively on the full 5,000-sample Danish training set and comparing evaluation metrics against both zero-shot and few-shot strategies.

3. Model Alignment with Human Annotations:

- (a) We hypothesize that including predictions from any of the trained XLM-R models alongside human annotations will not significantly increase inter-annotator disagreement. This will be assessed using changes in Mean Squared Difference and Krippendorff's Alpha, benchmarked against human-only annotations with bootstrapped confidence intervals.
- (b) We predict that all trained XLM-R models will exhibit annotator bias, tending to predict mid-range (average) levels of educational value. This will be investigated using confusion matrices and bootstrapped confidence intervals of global mean differences.

Roadmap This report is structured to systematically address the research questions and hypotheses. chapter 2, "**Data & Preprocessing**", covers the construction and characteristics of our datasets. chapter 3, "**Methods**", outlines our human annotation procedures, model architecture, and evaluation frameworks. chapter 4, "**Results**", presents the empirical findings from both human and automated annotation experiments. Finally, chapter 5, "**Discussion**", interprets these results, contextualizes them within our hypotheses, and suggests future research directions. For detailed navigation, each chapter includes a brief overview of its contents. Additional information is provided in the Appendices.

2 Data & Preprocessing

The success of any machine learning model is fundamentally tied to the quality of its data. For this project, which involves supervised fine-tuning and cross-lingual evaluation, a carefully constructed dataset was a central part of the methodology. This chapter details the multi-step process of sourcing, annotating, and balancing the data used for training and evaluating our educational content classifier.

We begin by introducing the foundational **FineWeb Family of Datasets** in section 2.1, from which all our data is derived. Next, section 2.2 details the complex process of **constructing our training datasets**, covering both the **English training set** (subsection 2.2.1) and the **Danish training set** (subsection 2.2.2). Following this, section 2.3 describes the **FineWeb-C human-annotated test set**, which serves as our "human standard" for evaluation. Finally, to ensure a consistent approach across the different datasets, section 2.5 explains how we addressed the **score scale mismatch** and created our universal scoring system, including how problematic, empty categories was handled (subsection 2.5.1). The underlying **assumption of equal interval values** for our ordinal scale is also discussed in section 2.4.

2.1 The FineWeb Family of Datasets

All data used in this project originates from the FineWeb family, a suite of large-scale, open-source datasets developed by the Hugging Face community [22]. The project's goal is to provide high-quality, extensively filtered web text for training and evaluating language models. The primary components we leverage are:

- **FineWeb:** The original, English-only version of the dataset [11]. It provides a massive, high-quality corpus of filtered web text sourced from Common Crawl. We use this, alongside its successor, as a source for unlabeled English text for our class balancing procedure.
- **FineWeb-2:** The successor to FineWeb, expanding the scope to be **multilingual** [12]. It is built from 96 Common Crawl dumps and contains nearly three trillion tokens, meticulously filtered for quality. This is the primary source for our unlabeled Danish documents.
- **FineWeb-Edu:** A subset of FineWeb where documents have been automatically classified for their educational value on a 0-5 scale [11]. The labels were generated by the `fineweb-edu-classifier`, a specialized BERT-like encoder model that was itself finetuned for this classification task. The dataset contains only text with an educational score of 3 or higher.
- **FineWeb-C:** A smaller, human-annotated subset of FineWeb-2 created specifically for benchmarking educational content classifiers. The "C" stands for "Community," as the labels are provided by human annotators using a 1-6 scale with an additional 'problematic content' label. [9] This dataset serves as our "human standard" test set for Danish.

The following sections will detail how we specifically processed and combined these components to construct our training and evaluation sets.

2.2 Training Data Construction

Our experimental design required two class-balanced training datasets of 5,000 samples each: one in English for zero-shot and few-shot transfer, and one in Danish for monolingual fine-tuning¹. As no such datasets were publicly available, we constructed them from the ground up using components from the FineWeb family, as described in section 2.1.

2.2.1 English Training Set: A Strategy for Class Balance

A robust classifier must be trained on a balanced set of examples from all classes. Our initial plan was to use samples from the FineWeb-Edu dataset. However, the dataset presented a major challenge: it was heavily skewed towards higher scores, for instance it containing only samples with an educational score of 3 or higher. Training on this data alone would produce a model biased against identifying non-educational or minimally educational content.

To solve this, we implemented a multi-step data augmentation and balancing process seen in Figure 2.1:

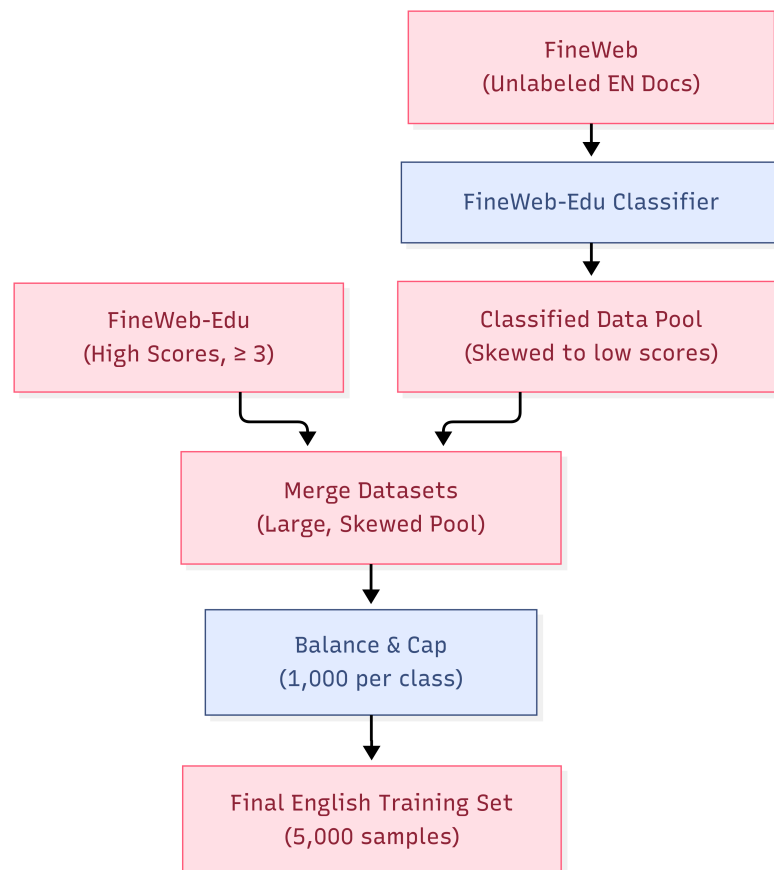


Figure 2.1: English pipeline. Blue boxes: Methods, Red: Datasets Unlabeled English text from FineWeb are first scored by the FineWeb-Edu-classifier, merged with high-score samples from the original FineWeb-Edu subset, and then class-balanced ($\leq 1\,000$ examples per score) to create a 5 000-sample balanced English training set.

1. **Automated Classification for Low-Score Samples:** We first took 10,000 unlabeled English documents from the general FineWeb dataset and classified them us-

¹Monolingual fine-tuning refers to training a model exclusively on labeled data from a single language, in our case, Danish

ing the official `fineweb-edu-classifier`. This allowed us to generate a large pool of samples with scores across the entire spectrum, particularly for the underrepresented low-score classes (0 and 1). A histogram of the resulting label distribution is shown in Appendix D

2. **Strategic Merging:** The newly classified data, while rich in low-score examples, was now skewed in the opposite direction. To achieve a uniform distribution, we created our final training set by merging samples from two sources: 8,000 samples from the original high-score samples from FineWeb-Edu and the 10,000 samples from the newly-classified low-score samples. To prevent data contamination and guarantee no duplicate texts, all samples were sourced distinct Common Crawl dumps. A complete list of the specific dumps used for the English training set is provided in Appendix C. A histogram of the resulting label distribution is shown in Appendix D
3. **Enforcing Balance via Capping:** Finally, we enforced a strict class balance by randomly sampling and capping the number of examples for each of the five universal scores at 1,000. This resulted in a perfectly balanced training set of 5,000 English samples, as illustrated in Appendix D figure D.2

This balancing procedure was critical, as the initial pool of data generated via automated classification was heavily skewed towards lower scores. Table 2.1 illustrates the dramatic shift in class distribution from our initial classified pool to the final, balanced training set used in our experiments. For a visual representation of the final balanced distribution, see the histogram in Appendix D

Table 2.1: Comparison of the class distribution in our large, automatically classified data pool versus the final, balanced English training set.

Score	Initial Approx. Count	Final Count (Balanced)
0 (None)	~550	1,000
1 (Minimal)	~3,000	1,000
2 (Basic)	~1,100	1,000
3 (Good)	~300	1,000
4 (Excellent)	~50	1,000
Total	5,000	5,000

2.2.2 Danish Training Set

For our Danish experiments, we needed a parallel 5,000-sample dataset with the same class balance. We considered two primary strategies:

1. **Synthetic Annotations:** Using the large generative decoder, model Gemini 2.5, to label unlabeled Danish text from FineWeb-2.
2. **Translation:** Translating labeled english data into Danish.

We initially explored synthetic annotations, using the prompt described in Appendix Figure B.1. However, this approach proved highly inefficient. Due to the natural rarity of high-quality educational material on the web, we would have needed to process a large number of text to find 1,000 samples for the "Excellent" and "None" category, resulting significant computational costs and API usage.

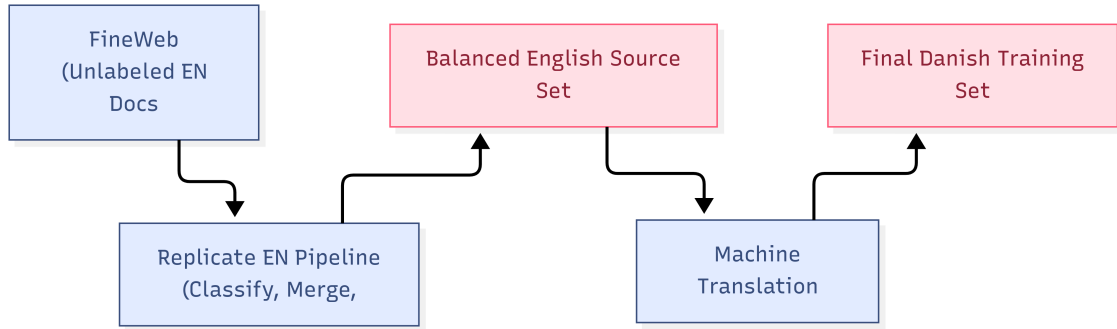


Figure 2.2: Danish pipeline. Blue boxes: Methods, Red: Datasets. The balanced English set is passed through the same preprocessing steps, then fully machine-translated (EN → DA) to yield a parallel 5 000-sample Danish training set with identical class distribution.

Consequently, we opted for a more efficient and precise method of translation, seen in Figure 2.2. A second, parallel English dataset was constructed to support the translation. This was done by following the exact data preprocessing pipeline described in subsection 2.2.1, but drawing the initial text from different web crawls to ensure our English and Danish training sets were completely distinct, see Appendix C. This 5,000-sample balanced English data set was then translated into Danish via the open source model "Helsinki-NLP/opus-mt-en-da".

This strategy offered a key advantage: it guaranteed a perfect one-to-one correspondence in class labels between the English and Danish sets, creating an ideal and controlled environment for studying cross-lingual transfer.

2.3 Test Data: The Fineweb-C Human Standard

To evaluate our models' performance, we required a test-dataset with high-quality annotations, that the models have never seen in training. For this purpose, we used Fineweb-C, the human-annotated benchmark for Danish educational content introduced in Section 2.1.

The dataset featured labels carefully created by a community of annotators, making it a reliable reference point. As part of this project, our team contributed by annotating 100 of these documents in accordance with the established guidelines (see Appendix A.1). This effort not only expanded the dataset but also provided the basis for the inter-annotator agreement analysis described in Section ??.

2.3.1 When in need of a "True" label

To perform a comparative analysis, and check the models performance, we need a test-set. An objective "True label" for how educational text is. How educational a text is, can vary from person to person. One person might find the text very educational, whereas another might find the same text ignorant.

In this paper, we assume that the mean of 6 annotaters, is the human standard label, or "True label". An example: $[0, 0, 0, 0, 3, 3] \Rightarrow \text{round}((0 + 0 + 0 + 0 + 3 + 3)/6) = 1$. This single integer is used as the "true" label in ??

2.4 Assumption of Equal Intervals

The ordinal ranks of the categories provide important information about agreement as the magnitude of disagreement must be larger between 'None' and 'Excellent' than 'None' and 'Minimal'. Therefore we aim to assign numeric scores for simplicity and consistency. However, this approach assumed equal intervals between categories, hence treating the categories as interval data. While this assumption may not perfectly reflect the true distances between categories, it introduced certain limitations in the generalizability of our results. Nonetheless, it was justified given the structured nature of the rating scale and its common interpretation in educational assessments such as the (0-5) score used in Fineweb-edu. [22]

2.5 Score: Scale Mismatch

The test data (Fineweb-C) and the training data employed different scoring systems for educational value. The Fineweb-C scale (C-scale) used **5 ranked labels** along with a distinct label for problematic content (see Appendix A.1). In contrast, the Fineweb-edu scale used in the training data (Edu-scale) employed **6 ranked labels** without a separate category for problematic content (Appendix B.1).

We first addressed the issue of the problematic content label, which appeared only in the C-scale. This label was absent from training data, source from FineWeb, due to prior filtering of problematic content during data preprocessing [11]. A manual review of a small sample of texts with the lowest educational scores revealed no clear instances of problematic content. Therefore, the “! Problematic !” label was excluded as a possible score. This exclusion also simplified the scoring by removing the only nominal category. As a result, all Fineweb-C examples annotated with “! Problematic content !” were filtered out, and the remaining annotations were assigned numeric scores (0–4) made possible by the assumption of equal intervals (Section 2.4).

Next, we considered the mismatch in the number of scoring categories. Although the Edu-scale included 6 ranked labels (0–5) and the C-scale only 5 (0–4), the extremes of both scales were well-aligned. A score of 5 on the Edu-scale, defined as “outstanding in its educational value,” closely matched the top C-scale score, “Outstanding teaching material.” This also held for the lowest scores, described as “No educational value” in the C-scale and as the absence of any relevant educational information in the Edu-scale.

This alignment suggested the feasibility of translating scores from the Edu-scale to the C-scale via a linear transformation, assuming equal intervals and aligned endpoints:

$$C_{\text{scale}} = \frac{4}{5}EDU_{\text{scale}}$$

Fineweb-C and Fineweb-edu used different scorings for educational value. The Fineweb-C scale (C-scale) employed **5 ranked labels** and problematic content label (Appendix: A.1), whereas the Fineweb-edu scale (Edu-scale) used **6 ranked labels** (Appendix: B.1) and did not handle problematic content separately.

First we considered the inclusion of the problematic content label, which was only present in the C-scale. The absence of the label in Fineweb-edu was due to a pre-filtering of the data. [ref]. Additionally, a manual examination of a small sample of texts with the lowest educational scores revealed no obvious instances of problematic content. Therefore, “! Problematic !” was excluded as a possible score. This decision also had the benefit of

discarding the only nominal category. Accordingly, all Fineweb-C data annotated with “! Problematic content !” were filtered out and we committed assigning the numeric scores (0–4) mentioned in section 2.4.

Secondly we considered the problem of different number of categories where the extremes of the two scales appeared to align: A score of 5 on the Edu-scale was only used when “the extract is outstanding in its educational value,” which corresponded closely to the C-scale’s top score, “Outstanding teaching material.” This alignment also held for the lowest scores, described as “No educational value” in the C-scale and as the absence of any relevant educational information in the Edu-scale. This suggested the possibility of translation between scales as a corrective preprocessing step, which made the assumption that the individually equal intervals and equal extremes between scales:

$$C_{\text{scale}} = \frac{4}{5} EDU_{\text{scale}}$$

While not ideal, this approach effectively reduced the bias introduced by using different scales. In pilot studies, the model often predicted approximately 25% higher than the average of Fineweb-C human annotations. This overprediction was notably reduced by applying the scale translation.

To maintain consistency with the evaluation set and human annotations, we adopted the C-scale as the standard for educational value throughout this work.

EDU-scale (0–5)	C-scale (0–5)
0 No educational criteria met	None (0)
1 “Basic information”	Minimal (1)
2 “Elements pertinent to education”	Basic (2)
3 “Appropriate for education”	Good (3)
4 “Substantial educational content”	Excellent (4)
5 “Outstanding in its educational value”	

Table 2.2: Comparison of model and community educational score labels

2.5.1 The Dead Category

An analysis of the training data revealed a notable heavy underrepresentation of the highest educational score. This caused the distribution to peak effectively around score 4. A similar pattern was observed in Fineweb-C annotations, where no example received a majority of “Excellent” annotations.

This scarcity of top-score examples effectively renders the highest category a “dead category”. One that exists in the label taxonomy but is unused in practice. As a result, models trained on this data fail to learn meaningful distinctions at the upper end of the scale, and rarely, if ever, predict the top score. This limitation must be acknowledged when interpreting model outputs and evaluating performance, particularly in scenarios that aim to surface highly educational content.

3 Methods

This chapter outlines the methodology used in our project, detailing the procedures and experimental design used to address our research questions and test our hypotheses, in section 1.2. Building upon the data collection and preparation described in chapter 2, we have structured this chapter into three main parts.

First, in section 3.1, we describe the **human annotation** process, covering the guidelines, bias mitigation strategies, and use of multiple annotators to ensure data quality.

Second, in section 3.2, we detail our **Model Architecture and Fine-tuning Strategy**, focusing on the XLM-R model, our regression-based fine-tuning, and cross-lingual training strategies.

Finally, in section 3.3, we present our **evaluation framework**, including our metrics for inter-annotator agreement and the **comparative statistical tests** (subsection 3.3.2) used to evaluate model performance. The results of these evaluations are presented in chapter 4.

3.1 Human Annotations

Annotations can be useful for training models, but they are inherently subjective, and it is easy to imagine examples with disagreement between annotators. To better understand the magnitude of this disagreement, we annotated 100 Danish datapoints from Fineweb-C [9]. For

To ensure compatibility with the dataset, we followed the established annotation guidelines used in Fineweb-C, where annotations aimed to capture the educational value of texts. The possible annotation categories can be examined in section 2.3.

When working with human annotations, addressing issues of bias, consistency, and reliability are crucial, especially in tasks involving subjective constructs such as educational value. In this project, all three authors annotated the same 100 danish texts from the FineWeb-C dataset. To aim for a high annotation quality, we considered the following before initializing the annotation process.

- **Annotation Guidelines:** Clear and consistent guidelines are essential for aligning annotators' understanding and reducing ambiguity [20][30]. We used the original FineWeb-C annotation guidelines to align with previous annotations. These guidelines were clear and included detailed explanations, criteria, and examples. (See Appendix A.1).
- **Minimizing Obvious Bias:** To avoid anchoring and conformity bias, we designed the annotation interface so that each annotator viewed only the raw text, without access to others' annotations. This setup aligns with standard practices for reducing social influence in labeling tasks [26, 21].
- **Multiple Annotators per Text:** Having multiple independent annotations per text improves reliability and helps capture variance in subjective interpretation. Each of the 100 texts was annotated by all three annotators and additionally had a minimum 3 previous annotations. This allowed us to measure inter-annotator agreement and assess ambiguity in the dataset which is presented in the following section.

During the annotation process, we observed that the quality of annotations may possibly decline due to fatigue or the influence of previously labeled items. Taking regular breaks proved important to maintaining objectivity and annotation consistency, in line with findings from [13] on annotator fatigue.

We acknowledge that, as annotators, we share similar backgrounds (particularly in terms of educational experience and cultural context—which) may influence how we assess educational value. Recent research has shown that such demographic factors can significantly shape annotation outcomes [17]. It is also likely that the original FineWeb-C annotators had comparable academic profiles, which may further limit the generalizability of the annotations. We address the implications of this limitation in the discussion section.

3.2 Model Architecture and Fine-tuning Strategy

In this section, we detail the architecture and methodology behind our automated approaches to classifying educational content. Our primary focus is on fine-tuning the **XLM-RoBERTa (XLM-R)** model, a lightweight encoder-based transformer, to specialize it for this task. Our methodology, was directly inspired by the FineWeb-Edu classifier framework [11].

3.2.1 Applying Fine-tuning to XLM-R

Large language models (LLMs) such as BERT, GPT, or XLM-R are typically pretrained on enormous collections of text using self-supervised learning objectives like masked language modeling. This pretraining phase enables the model to learn a general understanding of language, syntax, semantics, and even some world knowledge, without any human-labeled data.

However, pretraining alone does not make a model useful for specific applications. A model trained to fill in missing words in Wikipedia is not automatically able to classify whether a text is educational. For that, the model must be adapted to a concrete task using labeled data, a process known as supervised fine-tuning. During fine-tuning, the model's parameters are adjusted to minimize a task-specific loss function, effectively teaching it to map its general language understanding to task-specific predictions.

In this project, we apply supervised fine-tuning to specialize the pretrained XLM-RoBERTa (XLM-R) model for our task: annotating Danish web text for educational content. Each text was assigned a score between 0 (“None”) and 4 (“Excellent”). While this was formally a classification task, the ordinal nature of the annotation makes it tempting to assume equal intervals and model it using regression to better capture the relative magnitude of errors.

To evaluate the effectiveness of different cross-lingual training strategies, we investigated how well XLM-R can transfer knowledge from English to Danish by setting up three distinct **supervision regimes**, as detailed in the section subsection 3.2.3.

3.2.2 Model Architecture

We use XLM-RoBERTa (XLM-R), a multilingual masked language model pretrained on text from over 100 languages that achieves state of the art performance on crosslingual tasks. It builds upon XLM [cite] and RoBERTa [cite], using masked language modeling (MLM) to learn robust and generalizable token embeddings, a method where random input tokens are masked, and the model learns to predict these tokens from surrounding context. This method was popularized by BERT [10]. Specifically, we use the `xlm-roberta-base` model from the Hugging Face `transformers` library [18]. The architecture includes:

- 12 Transformer layers

- 768 hidden units per layer
- 12 attention heads
- 270 million parameters

Tokenizer XLM-R uses a shared vocabulary of 250,000 subword units¹ created by the SentencePiece tokenizer [16]. Unlike traditional tokenizers that rely on language-specific preprocessing (e.g., whitespace splitting or word segmentation), SentencePiece operates directly on raw text using a purely statistical, language-independent approach. This is particularly important for multilingual models like XLM-R, as it ensures consistent tokenization across all supported languages.

SentencePiece implements a subword segmentation algorithm such as Unigram Language Modeling [15] or Byte-Pair Encoding (BPE) [25]. These methods decompose words into smaller units (e.g., “education” → “educ”, “ation”) based on frequency patterns in the training corpus. This allows the model to:

- Efficiently handle rare or unseen words by recombining known subword fragments.
- Share vocabulary across languages, promoting cross-lingual generalization.
- Reduce out-of-vocabulary (OOV) issues and improve robustness.

For example, consider the English word *programming* and its Danish counterpart *programmering*. SentencePiece might tokenize them as:

```
English: ["_pro", "gram", "ming"]
Danish:  ["_pro", "gram", "mer" "ing"]
```

The leading underscore (_) denotes a word boundary. These shared subword components allow the model to align semantic patterns across languages and build multilingual representations in a consistent token space.

Architecture Modification For our fine-tuning task, we modify the standard XLM-R architecture by freezing the parameters of the main encoder body and appending a lightweight, trainable regression head.

- **Frozen Encoder:** By freezing the 12-layer XLM-R encoder, we preserve the powerful, general-purpose multilingual representations learned during its extensive pre-training. This prevents these representations from being drastically changed by our small, task-specific dataset, a phenomenon known as *catastrophic forgetting* [19]. This strategy significantly reduces the number of trainable parameters (from 270M to only a few thousand), which lowers computational costs and, crucially, mitigates the risk of overfitting.
- **Regression Head:** The head itself is a single linear layer that takes the final hidden state of the [CLS] token (a 768-dimension vector that summarizes the input sequence) and projects it to a single scalar prediction, $\hat{y}_i \in \mathbb{R}$. Only the weights of this small layer are updated during training.

Regression-based Modeling and Loss Function

Although this is technically a classification problem, our labels are ordinal, meaning they have a natural order where the distance between categories matters. For instance, misclassifying a '3' (Good) as a '4' (Excellent) is a much smaller error than misclassifying it as

¹

a '0' (None). Traditional classification methods, using a softmax layer and cross-entropy loss, ignore this crucial ordinal structure by treating all misclassifications as equally severe.

To properly capture this ordinal relationship, we shift from a classification framework to a regression problem. This approach is directly motivated by our prior assumption of equal intervals between categories (section 2.4), which allows the model to learn the relative distances between labels, not just the boundaries between them.

Consequently, we train the model to predict a single continuous value $y_i \in \mathbb{R}$ for each input text and use Mean Squared Error (MSE) as the loss function.

$$\mathcal{L}_{\text{MSE}} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3.1)$$

where:

- n is the number of examples in a batch,
- y_i is the true continuous score (e.g., 0.0, 1.0, ..., 4.0) for the i -th example,
- $\hat{y}_i$ is the continuous scalar value predicted by the model for the i -th example.

This setup ensures that larger mistakes are penalized more than smaller ones. This fits well with our task, since the labels are ordinal and not just categories. By using MSE, the model learns not only which class a text belongs to, but also how close or far off its predictions are on a continuous scale.

At inference/evaluation time, we convert the model's continuous prediction back in to class labels. This post-processing step is described in more detail in section 3.3

3.2.3 Cross-Lingual Transfer

Cross-lingual transfer refers to leveraging a model trained in one language (typically a high-resource language like English) to perform tasks in another language (such as Danish), often with minimal or no supervision in the target language.

Multilingual models like XLM-R make this possible by learning a shared interlingual representation space during pretraining [reference]. Using masked language modeling on large multilingual corpora, the model learns contextual embeddings that generalize across languages. As a result, linguistic features learned in English (e.g., structure, clarity, domain-specific terminology) can help predict educational value in Danish, even if the model has never seen Danish examples during fine-tuning.

This motivates our three regimes, visualized in Figure 3.1:

- **Zero-shot transfer:** The model is trained exclusively on English-labeled data and directly evaluated on Danish samples. This serves as our baseline for assessing cross-lingual generalization. This checkpoint also acts as the initialization point for the few-shot experiments.
- **Few-shot transfer:** Building upon the English-trained zero-shot model, we fine-tuned the model on a small number of Danish-labeled samples. This setup tests how well English training transfers to Danish with limited supervision. We investigate several few-shot settings with varying amounts of Danish data.

- **Full Danish fine-tuning:** Trained and evaluated exclusively on Danish data. Crucially, this model is initialized from the base XLM-R encoder and not from the English-trained model, thereby serving as a baseline.

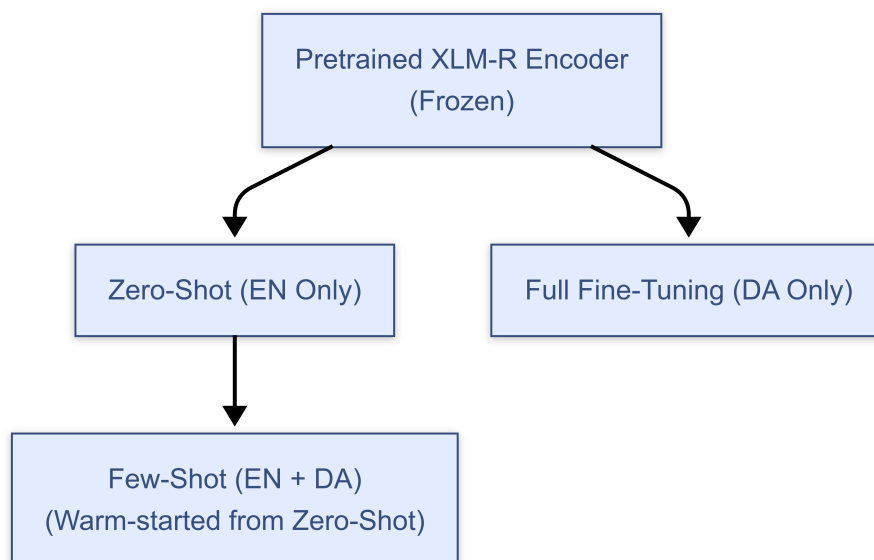


Figure 3.1: Overview of the three fine-tuning strategies used to train regression heads on top of a frozen XLM-R encoder.

3.2.4 Training, Validation and Model Selection

To ensure a fair comparison across supervision regimes, all models were trained using a the same optimization strategy and training configuration. A complete complete list of the parameters is available in Appendix E.

Optimization and Hyperparameters

The core of our training setup involved fine-tuning a regression head, which is a single linear layer, on top of a frozen XLM-R encoder to predict a scalar educational score $\hat{y}_i \in \mathbb{R}$.

For optimization, we utilized the AdamW optimizer. The learning rate was set to 1×10^{-5} and was managed with a linear warmup schedule, where the learning rate linearly increased from a very small value to its initial maximum over the first 10% of total training steps. This warmup phase is crucial for stabilizing training in the early stages, especially for transformer-based models. Training was conducted for 5 epochs with a batch size of 16, using Mean Squared Error (MSE) as the loss function to account for the ordinal nature of our labels.

Training Data Configurations: Each fine-tuning regime used a specific data configuration, designed to evaluate the various cross-lingual transfer strategies:

- **Zero-shot Training Data:** 5,000 English-labeled samples (as described in section 2.1 and subsection 2.2.1).
- **Few-shot Training Data:** Warm-started² from the English-trained zero-shot model, then fine-tuned with additional Danish samples (250, 1000, or 2500 samples) as detailed in subsection 2.2.2.

²Initializing a model's parameters with weights from a previously trained model

- **Full Danish Fine-tuning Data:** Initialized from the base XLM-R encoder and fine-tuned solely on 5,000 Danish-labeled samples, as constructed in subsection 2.2.2.

Hardware and Justification

All models were trained on a single Apple Silicon Mac, utilizing the *Metal Performance Shaders* (MPS) [3] framework for GPU acceleration and mixed precision. Training was implemented using the Hugging Face `transformers` and `accelerate` libraries.

We trained for 5 epochs. This choice was guided by two concerns:

1. **Preventing overfitting:** With small datasets and a powerful pretrained encoder, the model can start to memorize the training data quickly. Empirically, we observed that validation loss often plateaued after 600 optimization steps, which confirmed our suspicion.
2. **Efficiency:** Shorter training runs allowed us to iterate quickly across different supervision regimes. Since the encoder was frozen, only the lightweight regression head was updated, and fewer epochs were sufficient for convergence.

Model Selection To monitor the learning process and select the best performing model, we implemented a validation strategy. During training, we periodically evaluated the model's Mean Squared Error (MSE) on a held-out validation set (10% of the training data) (see Appendix E for specifics). The model checkpoint with the lowest validation MSE across all training steps was saved and used for the final evaluation on the test set.

This ensured that we selected the model at its peak generalization performance, rather than the one at the end of training, which might have started to overfit. The training loss curves for all trained models are presented in Figure 3.2.

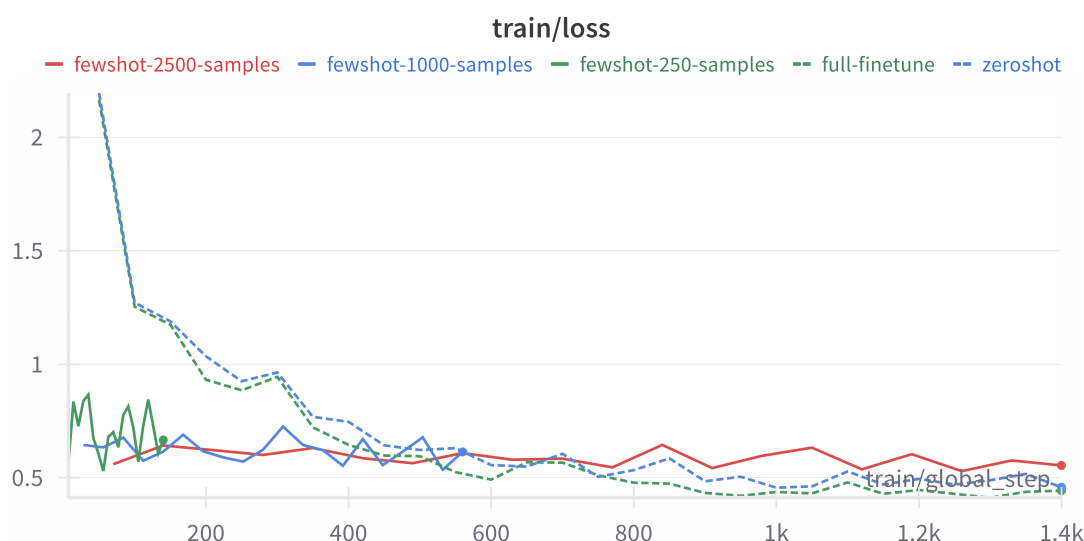


Figure 3.2: Loss function plotted over the training steps for the different models

When looking at figure Figure 3.2, it is important to keep in mind, that the zero-shot and full danish model are finetuned on "XLM-RoBERTa", the fewshot models are finetuned on the *zeroshot* model; the model has already seen 5000 samples, hence the low starting points.

Software and Code All models were trained using Python 3.11.5 and relied on the Hugging Face `transformers` library. The full source code, including data preprocessing, model training, and evaluation scripts, is available at: <https://github.com/DavidLindahl/danish-edu-llm-classifier> to ensure full reproducibility of our experiments.

3.3 Evaluation Framework

To evaluate our models and understand the nature of the annotation task, we designed a comprehensive evaluation framework. This section outlines the three core components of this framework: the metrics used to measure model performance, the metrics for assessing inter-annotator agreement, and the statistical tests used to determine if observed differences are significant.

Model Performance Metrics

To evaluate the performance of our fine-tuned models, we first need to convert their continuous regression outputs into discrete class labels. To accomplish this we clipped the raw model output between $[0, 4]$ and rounded to the nearest integer. The entire model inference pipeline is visualized in Figure 3.3

With the discrete labels, we evaluated our models using standard quantitative metrics such as: *accuracy*, *F1-score* and *mean-squared error (MSE)* complemented by *confusion matrix*, for class-wise error analysis.

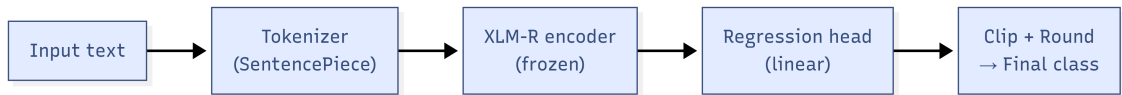


Figure 3.3: Inference pipeline for predicting educational content. The input text is tokenized using SentencePiece, encoded by a frozen XLM-R model, passed through a regression head, and finally mapped to a discrete class label via clipping and rounding of the scalar output $\hat{y} \in \mathbb{R}$.

This allows us to evaluate the regression-based model using discrete classification and disagreement metrics, as described below.

3.3.1 Inter-Annotator Agreement Metrics

Since the “true” category of a given text was unknown, it was natural to focus on measuring agreement between annotators, including in some cases, between human annotators and model predictions. To do this, we required a metric that could effectively capture the degree of agreement or disagreement between the scores assigned by multiple annotators to the same text.

Mean Squared Difference (MSD) Given our assumption of equal intervals between categories (see section 2.4), the MSD was a natural starting point. MSD quantifies disagreement by applying a squared difference function, which catch the magnitude of disagreement. Following [14], the pairwise difference function is defined as:

$$\delta^2(a, b) = (a - b)^2$$

Where a and b could be two annotation scores.

So to calculate the mean squared difference for a text, we considered all unordered distinct pairs, meaning (a, b) was considered equal to (b, a) . Thereby:

The mean squared difference for text i was:

$$MSD_i = \frac{2}{c_i(c_i - 1)} \sum_{1 \leq j < k \leq c_i} \delta(x_{ij}, x_{ik})$$

where x_{ij} denotes annotation j for text i and c_i is the number of coders (annotators) for text i . So to calculate the MSD across all texts, we simply calculated the average MSD_i over all texts, which provides the item-wise MSD:

$$MSD = \frac{1}{N} \sum_{i=1}^N MSD_i$$

where N is the total number of texts.

Krippendorff's Alpha (α) MSD expected total agreement as a baseline, which made it directly measure disagreement. However, it may be more interesting to consider a baseline of chance (or expected) disagreement to investigate text ambiguity more closely and better be able to compare it to literature. To do this we used Krippendorff's α [4]. Krippendorff's α is an agreement measure that can handle multiple annotators. For simplicity we will keep the assumption of equal intervals and use the δ -function from above. It differs from MSD by being a measure of agreement and introducing a normalization term of change disagreement:

$$\alpha = 1 - \frac{D_o}{D_e}$$

D_o is the observed disagreement (simplified for unordered distinct pairs) and turns out to be equal to our text-wise MSD measure. D_e is the expected disagreement, a term that uses the global distribution of annotations to calculate the change disagreement: [14].

$$D_e = \frac{2}{n(n-1)} \sum_{0 \leq a < b \leq R} n_a \cdot n_b \cdot \delta^2(a, b)$$

Where $R = 5$ is the number of ranks, n_a is the total number of times label a was assigned and n is the total number of annotations.

Thus, α yields a coefficient that ranges between -1 and 1 , where 1 signifies perfect agreement and 0 signifies chance agreement.

Annotator-Wise Agreement We used Krippendorff's α and MSD to assess overall disagreement among annotators. Additionally, we measured the disagreement of different groups of annotators to identify differences in consistency.

Category-Wise Agreement We were also interested in investigating the category-wise agreement. However, since we didn't know the true annotation for the texts, we used a pooled approach. For each educational-value, we created a pool of all the texts that included this annotation. This gave us 5 overlapping pools that we used to compute the same item-wise agreement measures. This was not a direct measure of the ambiguity of a category, but it indicated this by comparing which categories were mostly included in

texts with high disagreement.

Furthermore, we included a nominal version of the Krippendorff's α simply by changing the δ -function, so that ignores the magnitude of disagreement: [14]:

$$\delta_{nominal} = \begin{cases} 0 & \text{if } a = b \\ 1 & \text{if } a \neq b \end{cases}$$

This was included to provide a picture of the ambiguity not influenced by the rank-placement of the category. For instance "Excellent (4)" has a larger possible magnitude of disagreement than "Basic (2)". However, this came of cause with the cost of not catching the different magnitude of disagreement, hence we used both measures to provide full picture of the category ambiguity.

3.3.2 Comparative statistics

Beyond these metrics, we employed statistical comparative tests to quantify the relationship in and between models and human annotators. If nothing else stated, we used the standard 95% significant level and a null-hypothesis of no difference.

Difference of Agreement between Human Annotators To compare the disagreement of author and Fineweb-C annotations, we compared MSD_i for each group across the same texts. This was done under the assumption that both groups has same change agreement given equal texts and number of ranks. We computed the Shapiro Wilk test [?] and couldn't assume normality in any (p-value < 0.05). Since the groups shared texts, we used the paired non-parametric Wilcoxon Signed-Rank Test [24]. This tests if the distribution of the differences $MSD_{author} - MSD_{fineweb}$ is symmetric around zero.

Model performance To determine whether the performance differences between models were significant, we used bootstrapped confidence intervals (CIs) of the models' MSEs [?].

Impact of Agreement The impact the models had as included among the human annotators was investigated statistically with Bootstrapped CIs. The CIs was based on item-wise MSD and Krippendorff's α . Krippendorff's α can not be calculated item-wise and was bootstrapped by resampling with replacement.

Bias To investigate if the models were biased, we compared the model predictions individually against the human mean annotations. Again bootstrapping was used create CIs. This method provided a global image of the bias observed in the confusion matrices and was used to indicate if a global bias was present, significantly.

With range of tests using these two setups we aimed to better answer our initial predictions??.

4 Results

In this chapter, we present the results of our experimental test setup. These findings are structured to directly address the research hypotheses formulated in section 1.2.

We begin by analyzing the outcomes of our human self-annotation efforts (Section 4.1), showing insights into the subjectivity in educational content labeling. From there we perform the comparative statistical analysis as described in subsection 3.3.2. Then, we report on the performance of the various XLM-R fine-tuning, assessing their capabilities in automatically classifying Danish educational text across different cross-lingual strategies, using the metrics and statistical tests described in section 3.3. At last, we present confusion matrices, and means of annotators, to show the inherent bias in the models.

4.1 Self-annotation Results

The results of the self-annotation are presented in the tables below. The results are build on 100 texts with 3 author annotations and 3 fineweb-C annotations pr. text. The total agreement across all classes and authors was $\alpha = 0.44$ (Krippendorff) and The author annotators show higher agreement than Fineweb-C annotators. Finally the category with highest disagreement was "Excellent (4)" according to all agreement measures (see Table 4.2).

Table 4.1: Agreement scores (Mean Squared Difference, Krippendorff's α) and mean educational value score across annotations made by the authors and from FineWeb-C. Lower MSD and higher α indicate better agreement.

Annotation Source	MSD↓	Krippendorff's α ↑	Mean Score
Authors only	0.97	0.56	1.11
FineWeb-C only	1.61	0.34	0.95
All annotations	<u>1.26</u>	<u>0.44</u>	<u>1.03</u>

To statistically support the observation that the authors exhibited higher agreement, we conducted a Wilcoxon signed-rank test [24] on **item-wise (MSD_i) between annotators**. The test yielded a **p-value of 0.002** (< 0.05), leading us to reject the null hypothesis of equal disagreement between the author annotations and the Fineweb-C annotations.

In addition, we compared the mean annotation scores from the two sources. A Wilcoxon test on the **item-wise mean scores** resulted in a **p-value of 0.02**, also below the significance threshold of 0.05. This indicates a statistically significant difference in annotation tendencies, with the authors generally assigning slightly higher educational scores on average.

Table 4.2: MSD and Krippendorff’s α scores by educational value level. Each row reflects agreement metrics computed over all text samples in which the corresponding category was assigned at least once. Krippendorff’s α is calculated using both interval (squared difference) and nominal δ^2 -functions. Lower MSD and higher α values indicate greater agreement.

Educational Value	MSD↓	α (interval)↑	α (nominal)↑
None (0)	1.20	0.20	0.079
Minimal (1)	1.35	0.32	0.085
Basic (2)	1.85	0.21	0.010
Good (3)	2.25	0.10	-0.031
Excellent (4)	2.53	-0.016	-0.025

4.2 Model performance

Table 4.3: Global performance on the test set (against the Human Standard). Best automated model scores in **bold**.

Model	MSE↓	Accuracy↑	F ₁ -macro↑	F ₁ -weighted↑
XLM-R Zero-shot	0.426	0.420	0.374	0.367
XLM-R Few-shot 250	0.538	0.390	0.378	0.302
XLM-R Few-shot 1 000	0.438	0.430	0.434	0.379
XLM-R Few-shot 2 500	0.468	0.420	0.426	0.369
XLM-R full Danish Fine-tune	0.358	0.470	0.499	0.467

Table 4.3 presents a overview of the performance metrics for all experimental setups on the FineWeb-C test set, as described in section 3.3. The model indicating the strongest performance is the XLM-R full Danish Fine-tune model compared against the Human Standard.

However, these initial observations lack statistical grounding. Therefore, bootstrapped Confidence Intervals (CI) were conducted of the MSE for each model, which is shown in subsection G.8.2. Analyzing the plot we observed that all CIs were overlapping meaning that no model was performing significantly different in terms of mean MSE against the test-set.

4.3 The models impact on agreement

To investigate if our models would work on par with human annotators, we calculated the change in (Δ) MSD and α between (1) human annotations alone and (2) human annotations combined with model predictions (see Table 4.4). The results indicated minimal change in agreement among all models. This was further supported by bootstrapped CIs, where all CIs overlap with the human-only agreement. (see section G.2 and subsection G.8.3).

Full Danish fine-tune model was the only model with a positive impact on agreement when change agreement was taken into account (Krippendorff’s α).

Table 4.4: Model comparison using global Mean Squared Difference (MSD) and Krippendorff’s α between model predictions and human annotations. The metrics reflect the change when adding model predictions to human annotations, compared to human annotations alone. Lower Δ MSD and higher $\Delta\alpha$ indicate better agreement with human judgments.

Model	Δ MSD $\downarrow$	Δ Krippendorff’s α $\uparrow$
XLM-R Zero-shot	-0.08898	-0.02027
XLM-R Few-shot 250	-0.05706	-0.02777
XLM-R Few-shot 1 000	-0.08567	-0.02070
XLM-R Few-shot 2 500	-0.07714	-0.02132
XLM-R Full Danish Fine-tune	-0.10853	0.00151

4.4 Bias

4.4.1 Confusion Matrices

Figure 4.1 presents the confusion matrices for the evaluated models and human base-lines. As observed, all encoder-based models consistently overestimate the educational content score, exhibiting a bias towards predicting values higher than the test label. This is further supported by the higher average predicted scores from these models, as detailed in Table 4.5. Specifically, fine-tuned XLM-R models show an over-prediction bias, averaging 1.39 against the human average of 1.07

Table 4.5: Average Educational Scores for Models and Annotators on the FineWeb-C Test Set. Scores represent the mean predicted/assigned value on a scale from 0-4.

Source	Average Score
XLM-R Zero-shot	1.37
XLM-R Full Danish Fine-tune	1.25
XLM-R Few-shot 250	1.51
XLM-R Few-shot 1000	1.39
XLM-R Few-shot 2500	1.44
<i>Human Overall Average</i>	1.07

To dive further into this discovery of bias, we conducted bootstrapped CIs of the differences between model predictions and mean human annotations for each model. This was done to check, if there was a *significant* bias.¹ This is shown in subsection G.8.1. Analyzing the plot we observed that none of the CIs were including 0, meaning that the null-hypothesis of no difference was rejected. In other words, All models was significantly over-prediction in comparison to the Human Standard.

¹We also attempted to perform a Wilcoxon paired t-test, but due to the test-data having multiple labels, implementing a test was too time-consuming.

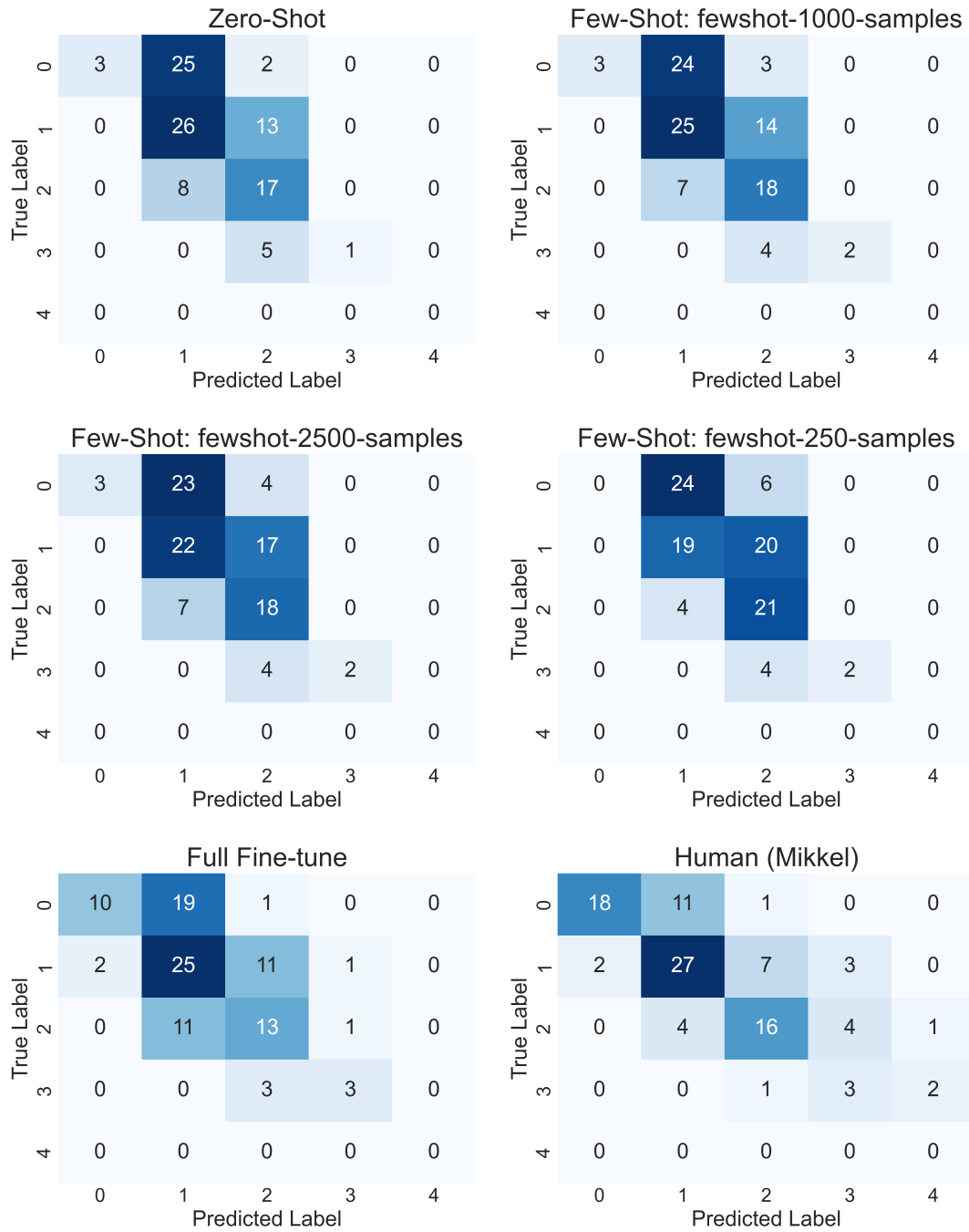


Figure 4.1: Confusion matrices on the FineWeb-C test set for: (a) XLM-R Zero-Shot model, (b) Few-Shot with 1000 danish samples, (c) Few-Shot with 2500 danish samples, (d) Few-Shot with 250 danish samples, (e) XLM-R Full Danish Fine-tune model, and (f) human reference (Mikkel). Darker cells indicate a higher count of instances. Rows represent test labels, and columns represent model predictions.

5 Discussion

5.1 Agreement among Human Annotations

Our results suggested relatively low agreement between the human annotators as the overall Krippendorff's α was $0.44 \ll 0.80$. 0.80 is considered the best candidate for a reliability threshold in [4]. The paper also mentions valid thresholds of 0.67 and 0.7; however, this is still far from our result. This result was worse than predicted in hypothesis 1,(a) (section 1.2)

However, it may be due in part to the subjective nature of this annotation task, where annotators have different views on educational value and interpret the guidelines differently. Furthermore, fatigue and lack of motivation and concentration may also play a role.

Examining some of the tests with high disagreement we often found the "None" annotation on texts with obvious educational value. An example is a text explaining a medical use of the "Jiaogulan" plant on horses. (found in appendix F. This is one of two examples where all five categories are present among the only 6 annotations.

5.1.1 Authors vs. Fineweb-C annotations

It was found that the authors had higher agreement than the Fineweb-C annotators supported by a p-value of 0.002 (< 0.05). (section 4.1)

This difference was however expected since the authors had high personal and professional similarities such as common education, age, gender, economic status and geography, which cannot be assumed for the Fineweb-C annotators. Furthermore we identified a methodological limitation in the annotation process: we didn't shuffle the order of the annotation task between each annotator, which could potentially introduce sequence-dependent biases (e.g., anchoring or fatigue effects). While the authors' observed moderate agreement ($\alpha=0.56$), this methodological choice represents a potential confounding factor. Weak tendencies can be observed in a graph of the mean author annotations across the text index, where sharp drops frequently follow highly rated texts (see Appendix Appendix H). To mitigate such potential biases and enhance data reliability in future annotation efforts, it is essential to enforce randomization of the task order.

Furthermore, the means of the author and Fineweb-C annotations were compared. We rejected the hypothesis of equal central tendency between the two groups, as indicated by a Wilcoxon test with a p-value of 0.02. This suggests that, while following the same annotation guidelines, the author annotations were systematically more generous, tending to assign higher educational scores than those in Fineweb-C. (see Table 4.5)

Despite adherence to the same annotation schema, subtle differences in interpretation could arise. Authors, being closely involved in model development and familiar with the educational framing of the task, may have been more attuned to content with implicit educational value. Furthermore, since the authors developed and refined the scoring framework, their internal consistency may have been higher.

Our results support **Hypothesis 1.b** (item 1), demonstrating significantly higher agreement among the authors compared to the Fineweb-C annotators, likely due to shared background and task familiarity.

The observation that 'Excellent (4)' was most prone to disagreement aligns with Hypothesis 1.3, which predicted greater ambiguity for content with higher perceived educational value.

5.1.2 Ambiguity across the categories of educational values

Investigating the classes of educational values with a pooled approach (see section 2.5) it appears that the annotation class "Excellent (4)" was mostly involved in texts with high annotation disagreement. Generally there seem to be a tendency that the higher the educational value, the higher the ambiguity. However, this tendency also follows the distribution of annotations and could be influenced by this.

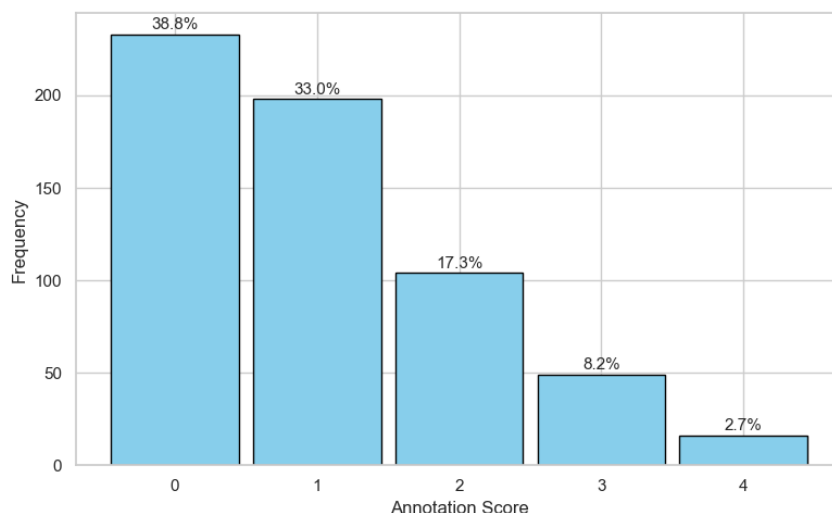


Figure 5.1: Distribution of all annotations across all texts

For comparison of categories the problem of class-imbalance increased since it resulted in large differences in pool size. This was feared to compromise the opportunity of fairly comparing categories.

Ambiguity measure. For further investigation it would be interesting to train another model to catch ambiguity. How uncertain are we about this data point.

The observed moderate human agreement **is crucial for understanding** the development and evaluation of automated annotation models. Since the 'true' educational value of a text can be subject to varied interpretation even among humans, the performance of an AI model trained on such data will inherently be constrained by this inter-annotator variability. This suggests that a model's 'accuracy' should be interpreted not as a measure of replicating a single, objective truth, but rather as its ability to align with the collective human judgment, including its inherent ambiguities. This human variability effectively sets a practical ceiling for automated performance on this task, underscoring the continuous need for refined annotation processes and potentially exploring methods to quantify or model this inherent ambiguity, to achieve a more robust 'ground truth' for such complex, subjective tasks.

5.2 Evaluation of Automated Models

The performance of our automated annotators, including the fine-tuned XLM-RoBERTa models, is summarized in Table 4.3.

A significant finding from our experiments is the strong cross-lingual transfer capability of the XLM-RoBERTa model. The zero-shot model, trained exclusively on English data and directly evaluated on Danish texts, achieved a commendable Mean Squared Error (MSE) of 0.556 and an F1-weighted score of 0.416. The overlapping bootstrapped confidence intervals (CIs) of the models' MSEs indicate no statistically significant difference in performance between them G.6. This demonstrates that even without exposure to Danish data during fine-tuning, XLM-RoBERTa's pre-trained multilingual representations enable it to effectively generalize to the "educational score" label across languages. This highlights the inherent strength of large multilingual foundation models.

While few-shot fine-tuning aims to improve performance with limited target-language data, our results showed a nuanced outcome. The **few-shot** models (250, 1000, and 2500 Danish samples) did not consistently outperform the zero-shot baseline on test set metrics. For instance, the fewshot-1000 model's MSE (0.566) was only marginally higher than the zero-shot, while the fewshot-250 (0.656) and fewshot-2500 (0.591) models exhibited slightly higher disagreement (worse MSE). This suggests that for a task where the multilingual model already provides a strong cross-lingual foundation, small increments of target-language data may not always yield proportional gains in generalization performance, or could potentially suffer from minor overfitting to the limited data.

In contrast, the **full Danish fine-tuning** model, trained on 5,000 Danish samples from scratch, achieved the best overall performance with an MSE of 0.477 and a F1-weighted score of 0.506. This improvement over the zero-shot and few-shot baselines indicates that a substantial amount of target-language data remains crucial for achieving optimal performance, even with highly capable multilingual pre-trained encoders. Although this result was not statistically significant (see Figure G.6), a larger test set may show a significant difference of performance.

Additionally, to provide a strong comparative baseline, we also evaluated the performance of Gemini 2.5 Flash, a powerful SOTA (state-of-the-art) generative decoder model, as an automated annotator (details in Appendix G.3). This allowed us to benchmark our fine-tuned XLM-R models against a highly capable, general-purpose LLM. The Gemini 2.5 Flash model performed comparatively, outperforming encoder models in some metrics, while being less biased, underlining the competitive performance of SOTA decoder models.

As illustrated in Figure 3.2, the loss trajectories for all fine-tuned few-shot models display minimal to no decrease throughout training. One possible explanation is that the zero-shot model had already reached a performance plateau for this particular regression task, potentially limiting the scope for further improvement. However, this interpretation remains speculative. Future work could explore whether these trends persist under alternative conditions, such as when using a monolingual rather than multilingual model architecture, or when increasing the number of training examples available during fine-tuning.

5.2.1 The Test Set

The models were evaluated against the Human Standard, as true labels are inherently unattainable in a subjective task such as annotating educational value (see section 2.3). This, of course, introduces limitations, particularly due to the high level of inter-annotator disagreement, as discussed in Section 5.1. Such variability in human judgment compromises the reliability and interpretability of the evaluation, making it difficult to draw strong

conclusions about model performance. To mitigate the influence of this human inconsistency and achieve more robust evaluations, future work could involve testing encoder models on a significantly larger test set.

5.2.2 Models Alongside Human Annotators

One way to address the absence of a definitive ground truth in subjective tasks is to examine how disagreement changes when models are included alongside human annotators. By incorporating model predictions into the pool of human annotations, we can assess whether the models behave similarly to humans in terms of variability and judgment.

As shown in Table 4.4, the overall disagreement increased slightly when model predictions were added. However, this increase was not statistically significant, suggesting that the models do not drastically deviate from the level of variability already present among human annotators.

This analysis, however, relies on the assumption that our human annotations offer a reasonable and representative benchmark for high-quality human judgment. To support this, we investigated inter-annotator agreement and patterns of consistency across annotators (see Section 5.1). While some variability is expected in subjective assessments, our findings revealed a high level of disagreement among human annotators, raising concerns about the reliability of the annotation process as a comparative standard.

Although models do introduce some additional disagreement, their inclusion does not appear to dramatically distort the overall annotation landscape. This supports the notion that large pre-trained models may be approaching human-level variability in complex subjective tasks like evaluating educational value.

5.3 Discussion of Bias

As highlighted in our results (section 4.4, specifically Table 4.5 and Figure 4.1), all encoder-based models consistently over-predicted the educational value of texts compared to the human standard. The average predicted score from our fine-tuned XLM-R models was 1.39, notably higher than the human overall average of 1.07. This pattern is visually confirmed in the confusion matrices (Figure 4.1), which illustrate a skew towards higher categories. As mentioned in section 4.4, there is a *significant* bias backed by the confidence intervals.

We suspected that the training data was biased towards having too high labels, so as described in section 2.5, we multiplied all training data labels with $\frac{4}{5}$. In hindsight, this action probably misaligned the model's decision boundaries. Future research should re-evaluate this scaling, as it may have inadvertently contributed to the observed bias.

Further analysis of the confusion matrices (Figure 4.1) directly reveals a pronounced tendency for the models to primarily predict categories '1' (Minimal) and '2' (Basic). Predictions for '0' (None) are notably sparse, and categories '3' (Good) and '4' (Excellent) are almost entirely ignored. For instance, the zero-shot model predicts category '3' only once and category '0' three times (see Figure 4.1a).

This behaviour points to, that we have trained some model that love predicting middle values, and is penalized too hard for predicting the extremes. This ties directly with the choice of loss function: MSE.

To further investigate this bias, we performed the *exact* same experiment but with the Class Density Weighted Cross-Entropy (CDW-CE) loss function. The confusion matrices and performance for these models can be seen in Appendix section G.5 and Table G.3. Table G.3.

It was observed that this alternative approach resulted in reduced bias compared to the MSE-trained models, which tended to cluster predictions predominantly around categories '1' and '2'. This supports the idea that MSE might penalize misclassifications for broader categories too heavily. The F1-macro metric was also higher across the board, and the models performed better, the more data they saw. This supports the idea that MSE might penalize misclassifications for broader categories too heavily.

However, even in this classifier, we see slight bias. A potential explanation for this persistent, albeit subtle, bias lies in the composition of our test data. With only 6 human annotators contributing to the human standard, inherent inconsistencies and the subjective nature of the task (as discussed in section 5.1 and the bias in test set seen in Figure 5.1) might limit the models' ability to achieve perfect unbiasedness against this reference.

5.4 Revisiting the assumption of equal intervals

Our current work, as detailed in section 3.1 and subsection 3.2.2, operates under the assumption of equal intervals for the ordinal annotation categories. While this simplifies the task and aligns with common practice for similar scales, it presents avenues for future research into more nuanced approaches.

One such avenue involves exploring alternative difference measures for computing disagreement. For instance, it is possible to obtain more accurate insights into inter-annotator disagreement by using a difference measure that explicitly respects the ordinal nature of the categories, rather than strictly assuming equal intervals [14]. An example of such a measure is the ordinal squared difference:

$$_{ordinal}\delta^2(a, b) = \left(\sum_{g=a}^b n_g - \frac{n_a + n_b}{2} \right)^2$$

Where n_a and n_b are the frequencies of the compared categories, and the $\sum n_g$ term includes frequencies of all categories between a and b , inclusive. Implementing such measures in future analyses could provide a deeper understanding of disagreement without being constrained by the equal interval assumption.

Furthermore, the choice of loss function for training the regression head is directly influenced by this assumption. While Mean Squared Error (MSE) was adopted for its simplicity and conceptual fit with the equal interval assumption, future work could explore using different loss-functions, that don't require interval data. Instead, one could investigate incorporating a loss function directly derived from this *ordinal squared distance* concept. This would involve penalizing errors based on the true ordinal relationship between predicted and true labels, where the "distance" is not fixed as in MSE but rather reflective of the non-uniform spacing or importance between categories as defined by an ordinal scale. Developing a loss function that takes this into account, could potentially take in more information on the training data, leading to a better model.

However, when comparing Krippendorff's α using ordinal and interval difference measures across all human annotations, the result was surprisingly similar — the absolute difference

was just 0.0037. This small gap raises the question of whether the more complex ordinal difference measure adds meaningful value in this case.

Furthermore, Ordinal measures are often praised for respecting the rank order of categories without assuming equal spacing, but they also depend heavily on the actual distribution of the data. In situations with strong class imbalance — like ours — this can distort the disagreement calculation. In contrast, the interval difference measure is simple and doesn't depend on how labels are distributed. So, If the assumption of equal spacing reflects the underlying construct (e.g., severity, quality, or agreement), then this approach not only simplifies the analysis but also embeds a reasonable prior about the data's structure. In that sense, it can contribute useful information rather than distorting it — as long as the assumption holds.

So, given how little the two measures differ here, and considering the potential downsides of relying on ordinal frequencies, it's reasonable to question whether the ordinal formulation is always the better choice. In many practical cases, the interval-based analysis might offer a more consistent and interpretable picture of annotator reliability.

5.5 Relevance: Pretraining data for decoder models

5.5.1 Why *Quality* Now Beats Sheer Token Count

As decoder LLMs approach trillions of parameters, simply adding more and more weights or blindly expanding the corpus, often just result in worse results.

Evidence now shows that the dominant failure modes, hallucinations, toxicity, and biased reasoning, is a consequence of low-quality “long tail” of web text introduced during pre-training [6, 7].

Put plainly: garbage in, garbage out. These studies therefore argue that, under a fixed compute budget, replacing noisy tokens with carefully chosen ones is a more powerful lever for capability and safety, rather than further scaling alone.

Together, these findings highlight our educational annotations as a practical solution to the data quality bottleneck. the following sections explore their impact and remaining challenges.

- referer til det som nickis pitch. Det som handlede om begrænset data på verdensplan

5.5.2 How relevant was this study for the future...

- Antag at AI kan løse alt i fremtiden

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A Fineweb-C: Guidelines for Rating Educational Content

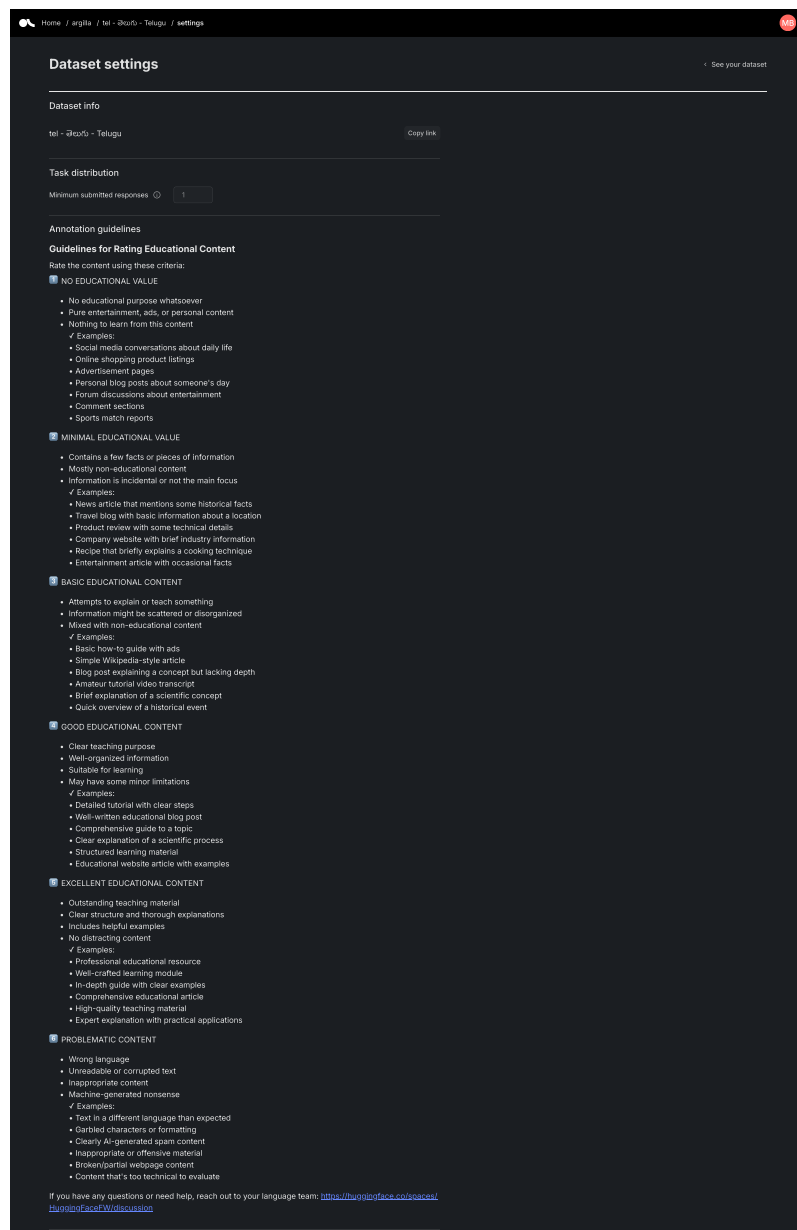


Figure A.1: Login is needed to see the guidelines on a webpage. LINK: Fineweb-C score guideline

1

B Fineweb-Edu: Prompt defining education value score

Below is an extract from a web page. Evaluate whether the page has a high educational value and could be useful in an educational setting for teaching from primary school to grade school levels using the additive 5-point scoring system described below. Points are accumulated based on the satisfaction of each criterion:

- Add 1 point if the extract provides some basic information relevant to educational topics, even if it includes some irrelevant or non-academic content like advertisements and promotional material.
- Add another point if the extract addresses certain elements pertinent to education but does not align closely with educational standards. It might mix educational content with non-educational material, offering a superficial overview of potentially useful topics, or presenting information in a disorganized manner and incoherent writing style.
- Award a third point if the extract is appropriate for educational use and introduces key concepts relevant to school curricula. It is coherent though it may not be comprehensive or could include some extraneous information. It may resemble an introductory section of a textbook or a basic tutorial that is suitable for learning but has notable limitations like treating concepts that are too complex for grade school students.
- Grant a fourth point if the extract is highly relevant and beneficial for educational purposes for a level not higher than grade school, exhibiting a clear and consistent writing style. It could be similar to a chapter from a textbook or a tutorial, offering substantial educational content, including exercises and solutions, with minimal irrelevant information, and the concepts aren't too advanced for grade school students. The content is coherent, focused, and valuable for structured learning.
- Bestow a fifth point if the extract is outstanding in its educational value, perfectly suited for teaching either at primary school or grade school. It follows detailed reasoning, the writing style is easy to follow and offers profound and thorough insights into the subject matter, devoid of any non-educational or complex content.

The extract: <extract>.

After examining the extract:

- Briefly justify your total score, up to 100 words.
- Conclude with the score using the format: "Educational score: <total points>."

Figure B.1: LINK: Fineweb-EDU score prompt-text

,

C Common Crawl Data Sources

To ensure the reproducibility of our experiments and prevent data contamination between our training and test sets, we carefully documented the specific Common Crawl dumps from which our data was sourced. All documents were drawn from the FineWeb and FineWeb-2 datasets, which are built upon these dumps. The following table details the exact sources for each component of our data.

Table C.1: List of Common Crawl dumps used for constructing the English and Danish datasets. The strict separation of sources prevents data leakage.

Purpose	Language	Common Crawl Dump ID(s)
English Training Set	English	
Fineweb		CC-MAIN-2024-10
Fineweb-edu		CC-MAIN-2024-22
Danish Training Set	Danish	
Fineweb		CC-MAIN-2022-40
Fineweb-edu		CC-MAIN-2023-06

D Label Distributions During Data Construction

This appendix documents the distribution of annotation scores during the construction of our English training set. We present three stages: raw classifier outputs, the merged dataset, and the final class-balanced version.

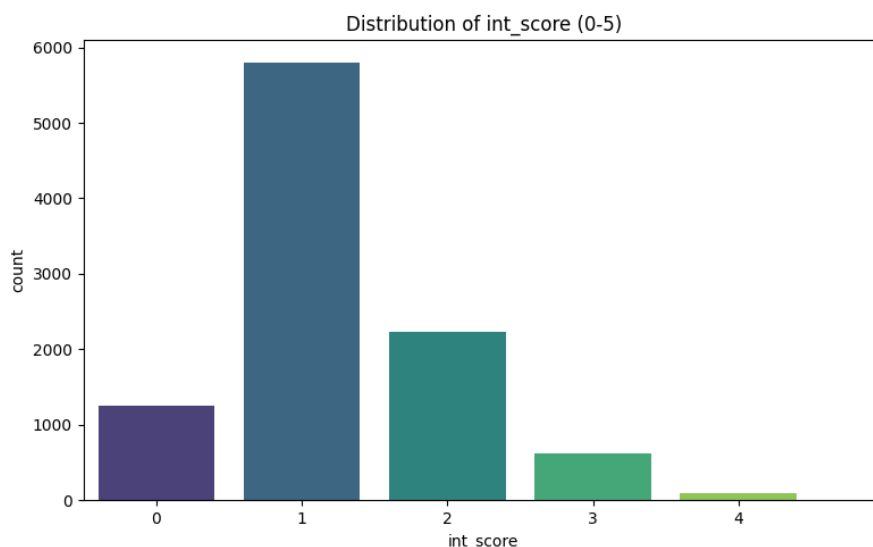


Figure D.1: Distribution of annotation scores predicted by the fineweb-edu-classifier on 10,000 unlabeled English documents. Low-score classes (0 and 1) dominate, revealing a skew toward minimally educational content.

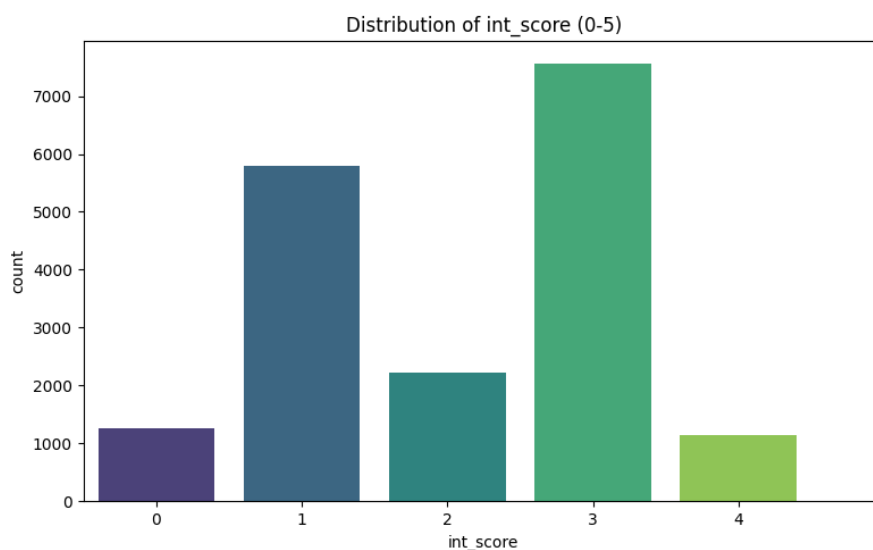


Figure D.2: Distribution of *int-score* after merging low-score samples, high-score FineWeb-Edu samples, and mid-score fillers. The merging step helped smooth the label distribution.

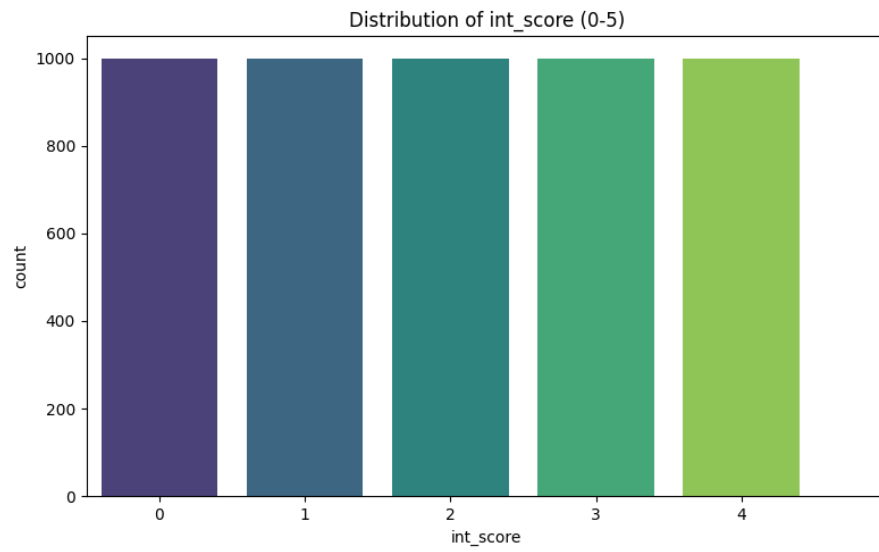


Figure D.3: Distribution of *int-score* after capping the merged dataset at 1.000 samples for each class.

E Hyperparameter Settings

[H] Training Parameters:

- - `learning_rate` = 1×10^{-5}
 - `num_train_epochs` = 5
 - `per_device_train_batch_size` = 16
 - `per_device_eval_batch_size` = 32
 - `validation_splits` = 0.1
 - `loss_function` = Mean Squared Error
 - `optimizer` = AdamW
 - `weight_decay` = 0.01
- **Learning Rate Scheduling:**
 - `lr_scheduler_type` = linear
 - `warmup_ratio` = 0.1
- **Evaluation and Logging:**
 - `eval_strategy` = steps
 - `save_strategy` = steps
 - `save_total_limit` = 2
 - `eval_steps` = 50
 - `logging_steps` = 50
 - `eval_on_start` = True
 - `load_best_model_at_end` = True
 - `metric_for_best_model` = eval_loss
- **Device & Miscellaneous:**
 - `use_mps_device` = True
 - `seed` = 42

F Text Snippet from Test Set With Human Annotated Scores

ID: 9c41aa98-225a-436f-8fb9-cc2b6588085f

Text (first 1000 signs):

Guzmania - Guzmania

Flerårige urteagtige stempløse planter, epifytter. Radikale blade, der danner en roset, lineær og xiphoid, hele. Peduncle dækket med små blade (forlader midten af roset af blade). Inflorescens spiciform eller panicle, nogle gange i form af et hoved. Blomsterne er placeret i axils af de røde bracts; corolla spongat, med segmenter kortere end røret; keglerne krydser i bunden. Generelle oplysninger om anlægget Guzmania - Guzmania Guzmania - Guzmania bromeliad familie. Antal: I en slægt omkring 110 arter. Oprindelsessted: Central- og Sydamerika. Anvendelse: blomstring. Plante størrelse: op til 30 cm i højden. Vækst: langsom. Blomstrende: om sommeren, efterår-vinter. Guzmania arter - Guzmania Guzmania lingulata var. Lingulata Epifytter. Planter 40-45 cm høje. Bladene er bælteformede, 45 cm lange og op til 6 cm brede, bredere ved bunden, grøn over, bunden med rødbrune linjer, bar. Blomstring er et tæt hoved. Kronblade hvidlig; skovle 7-10 cm lang, skarlagensrød. Colombia, Me...

Scores: [4, 3, 2, 0, 0, 1]

ID: c42e264c-ccdf-426c-bff3-265ed9a567e3

Text (first 1000 signs):

Beskrivelse

Jiaogulan er en plante der stammer fra østen, som er kendt for at understøtte den metaboliske hest. Den kan også gavne blodgennemstrømning i hovene, respiration, immunrespons, muskelfunktion, bindevæv og tarmfunktion. - Understøtter den metaboliske hest, uanset om det er på et tidligt eller sent stadie i tilstanden - Kan medvirke til at forbedre insulinfølsomheden - Stimulerer blodgennemstrømning i hovene - Kan afhjælpe øget puls i benene pga ømme hove - Til heste med hovsenebensproblemer - Understøtter sundt bindevæv - Kan medvirke til at understøtte hestens respirationsfunktion - Kan medvirke til at understøtte sund tarmfunktion ... Det anbefales at give hesten ½ – 2 g. pr. 100 kg kropsvægt pr. dag og fordele det på min. 2 udfordringer. For at få den mest gavnlige effekt bør den udfodres enten 20 minutter før resten af krybbefoderet, eller som minimum 2 timer efter resten af krybbefoderet (kan evt blandes med en...

Scores: [3, 4, 1, 3, 2, 0]

G Additional Results And Findings

G.1 P-values of difference between Model MSE

	Zeroshot	Full-finetune	Fewshot-250	Fewshot-1000	Fewshot-2500
Zeroshot	—	1.3×10^{-4}	4.5×10^{-12}	2.4×10^{-11}	1.9×10^{-12}
Full-finetune	1.3×10^{-4}	—	3.8×10^{-1}	1.2×10^{-3}	2.1×10^{-1}
Fewshot-250	4.5×10^{-12}	3.8×10^{-1}	—	4.9×10^{-12}	2.1×10^{-10}
Fewshot-1000	2.4×10^{-11}	1.2×10^{-3}	4.9×10^{-12}	—	7.0×10^{-13}
Fewshot-2500	1.9×10^{-12}	2.1×10^{-1}	2.1×10^{-10}	7.0×10^{-13}	—

Table G.1: Wilcoxon p-values between regression models. Bold values ($p > 0.05$) indicate non-significant differences.

G.2 The bootstrapped CI of mean MSDs of human annotations + each models predictions separately

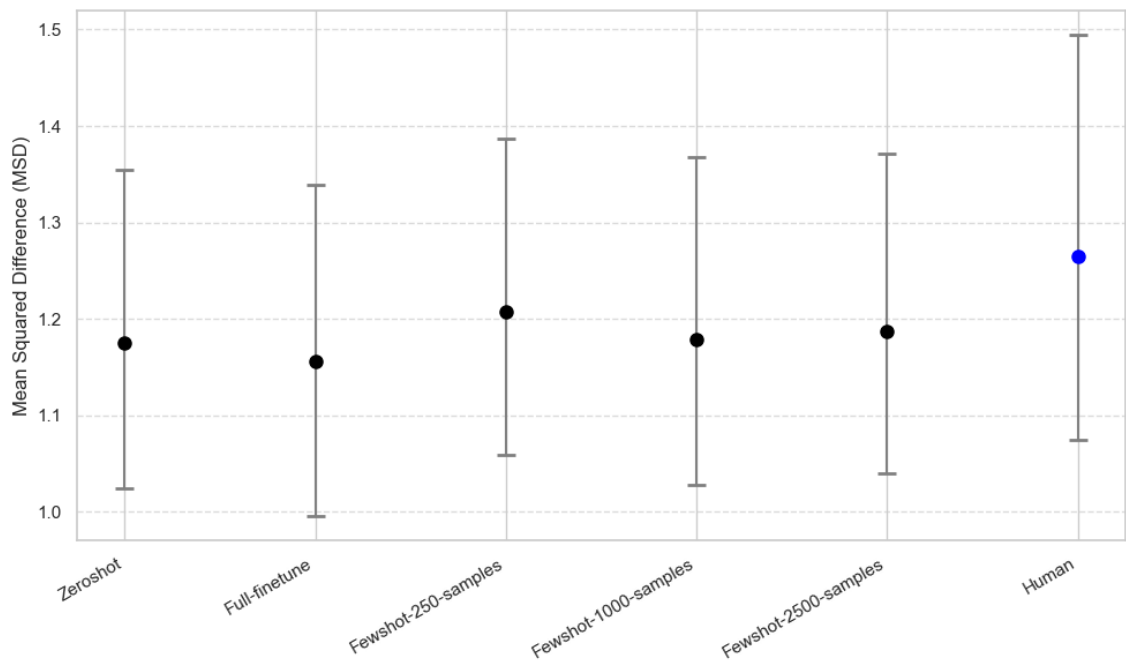


Figure G.1: The bootstrapped CI of mean MSDs of human annotations + each models predictions separately. 10000 resamples. Alpha = 0.05

Table G.2: Global performance on the testset. Best automated model scores in **bold**. Includes the autoregressive gemini-2.5-falsh transformer decoder for comparisons

Model	MSE↓	Accuracy↑	F ₁ -macro↑	F ₁ -weighted↑
Gemini 2.5 Flash	0.810	0.560	0.409	0.541
XLM-R Zero-shot	0.556	0.470	0.387	0.416
XLM-R Few-shot 250	0.656	0.420	0.374	0.341
XLM-R Few-shot 1 000	0.566	0.480	0.443	0.430
XLM-R Few-shot 2 500	0.591	0.450	0.424	0.406
XLM-R full Danish Fine-tune	0.477	0.510	0.511	0.506
<i>Human (David)</i>	0.460	0.600	0.468	0.591
<i>Human (Mikkel)</i>	0.510	0.640	0.480	0.659
<i>Human (Carl)</i>	0.768	0.526	0.427	0.514

G.3 Results including Gemini-2.5-flash for comparison

G.4 Score Distribution by Source

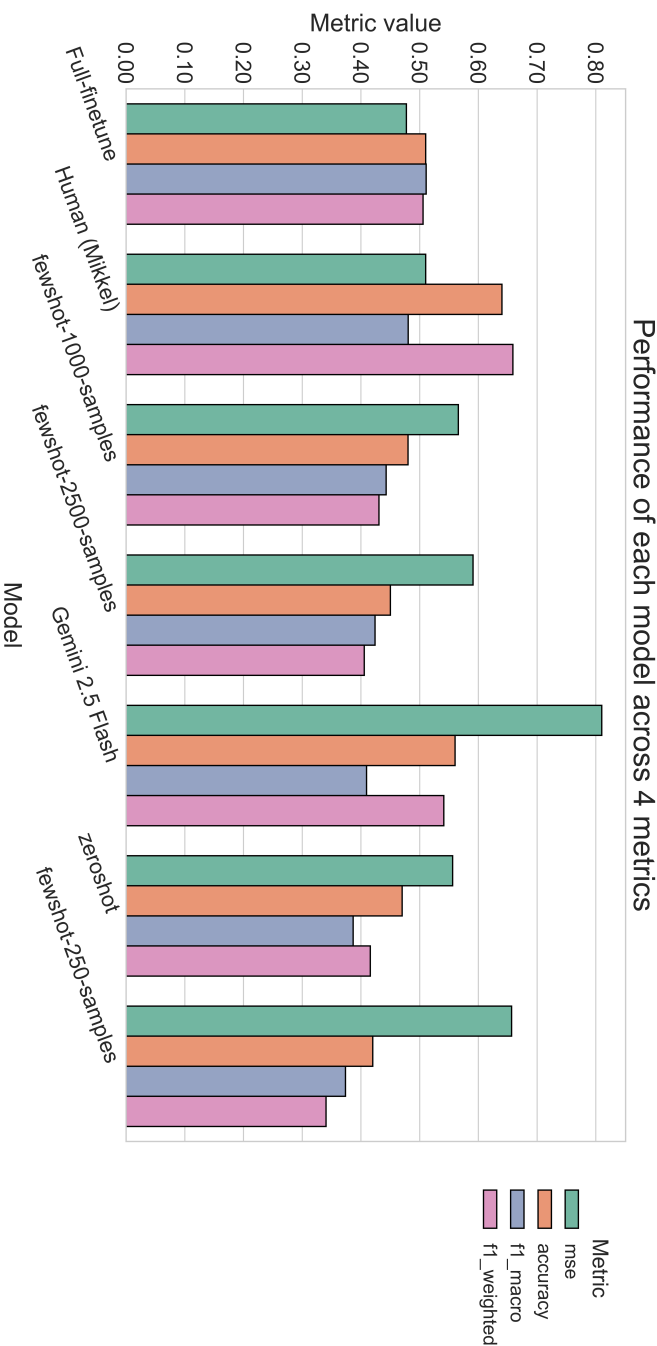


Figure G.2: Performance of all models across four metrics: Mean Squared Error (MSE), Accuracy, F1-macro, and F1-weighted, evaluated on the FineWeb-C test set. The full Danish fine-tune model performs best overall, closely followed by human annotators. Zero-shot and few-shot models show reduced but comparable performance, while Gemini 2.5 Flash demonstrates strong accuracy but high MSE due to underestimation.

G.5 Comparative Model Performance Across Evaluation Metrics

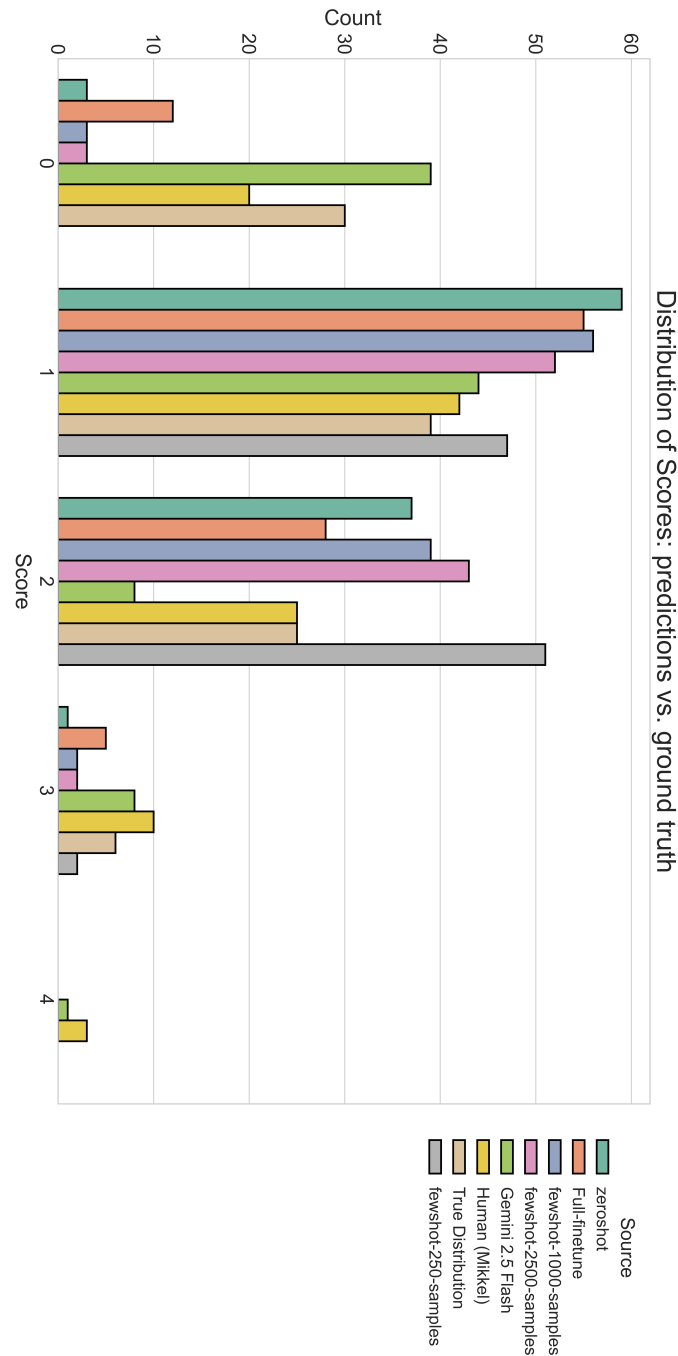


Figure G.3: Distribution of predicted and true educational scores across all evaluated models and human annotators. While human and high-performing models align reasonably well with the ground truth (gray bars), models like the few-shot 250-sample and zero-shot variants display skewed distributions. The Gemini 2.5 Flash model underpredicts, while others tend to overpredict scores.

G.6 Classification confusion metrics

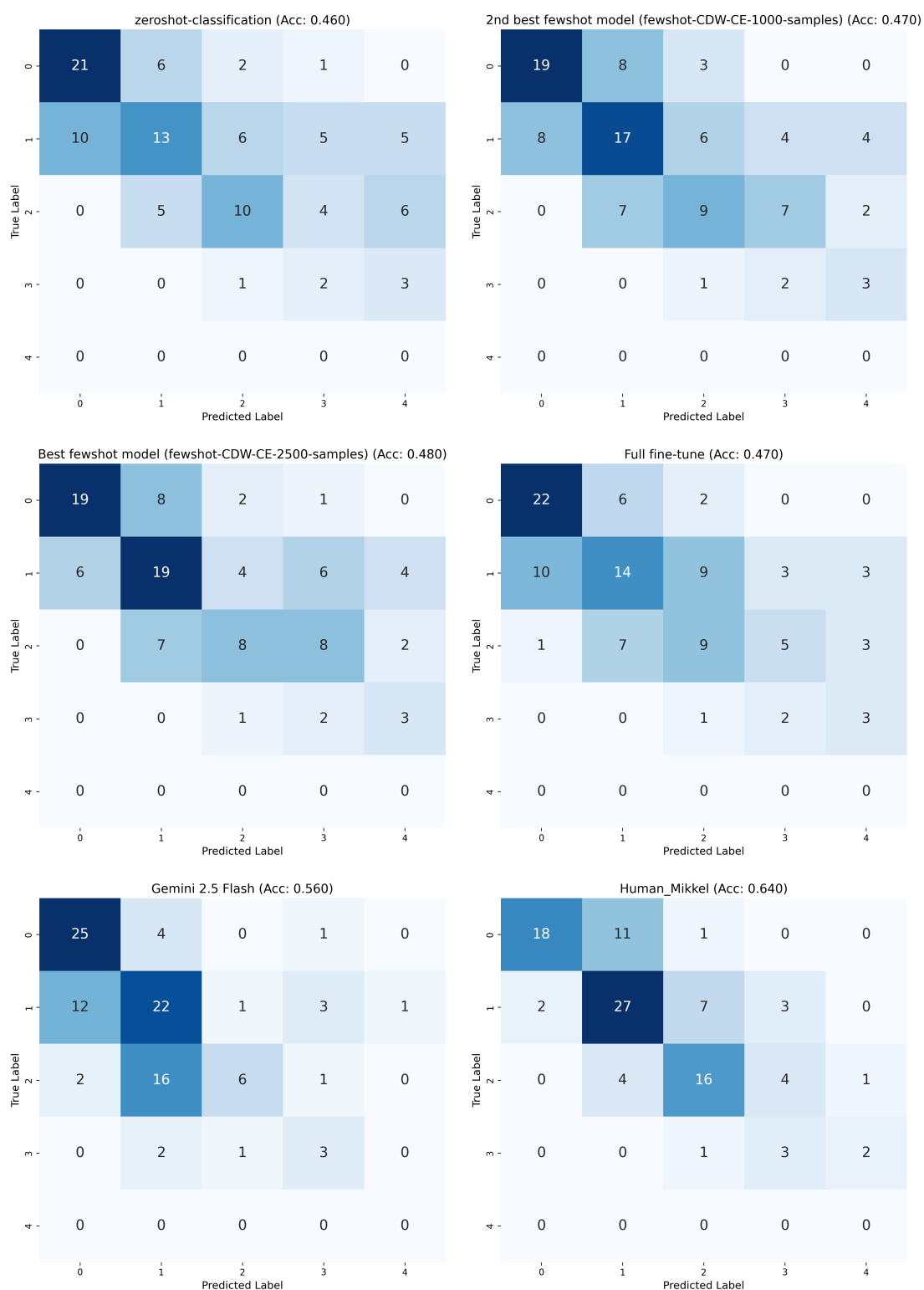


Figure G.4: Confusion matrices for classification models on the FineWeb-C test set. Darker cells indicate a higher count of instances. Rows represent true labels, and columns represent model predictions.

Table G.3: Global performance on the test set for classification models.

Model	MSE↓	Accuracy↑	F₁-macro↑	F₁-weighted↑
Gemini 2.5 Flash	0.810	0.560	0.409	0.541
XLM-R Zero-shot	0.556	0.470	0.387	0.416
XLM-R Few-shot 250	0.656	0.420	0.374	0.341
XLM-R Few-shot 1 000	0.566	0.480	0.443	0.430
XLM-R Few-shot 2 500	0.591	0.450	0.424	0.406
XLM-R full Danish Fine-tune	0.477	0.510	0.511	0.506
XLM-R Zero-shot (CDW-CE)	1.410	0.460	0.356	0.495
XLM-R Few-shot 250 (CDW-CE)	1.410	0.460	0.356	0.495
XLM-R Few-shot 1 000 (CDW-CE)	1.120	0.470	0.353	0.502
XLM-R Few-shot 2 500 (CDW-CE)	1.220	0.480	0.357	0.521
XLM-R Full Danish fine-tune (CDW-CE)	1.040	0.470	0.353	0.488
<i>Human (David)</i>	0.460	0.600	0.468	0.591
<i>Human (Mikkel)</i>	0.510	0.640	0.480	0.659
<i>Human (Carl)</i>	0.768	0.526	0.427	0.514

G.7 Global performance metrics for classification models

G.8 Bootstrapped CI

G.8.1 Bootstrapped CI to check bias

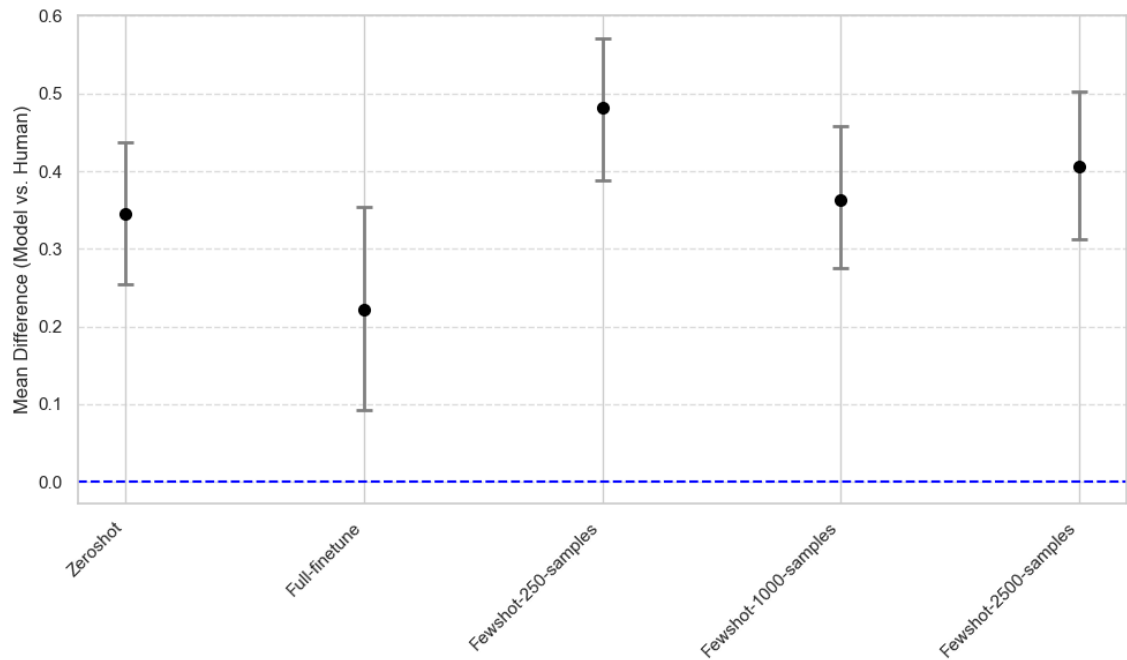


Figure G.5: Bootstrapped CI of the differences between model predictions and Human Standard (item-wise mean of human annotations). The bootstraps was created with 10,000 resamples and a confidence level of 95%.

G.8.2 Bootstrapped CI of Average MSE

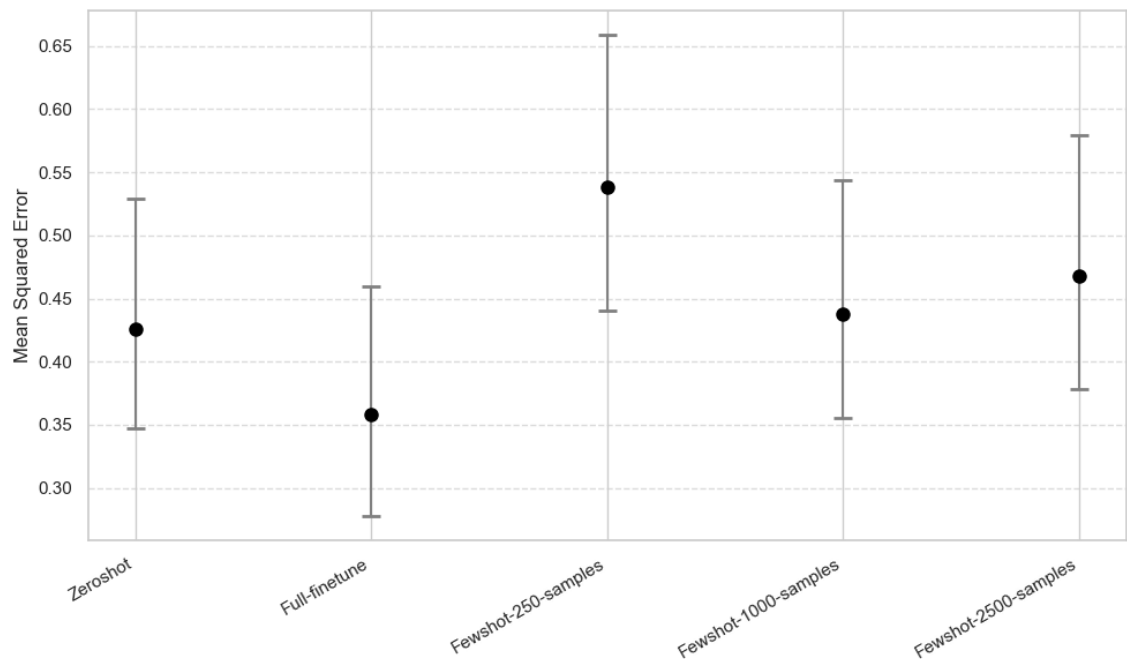


Figure G.6: Bootstrapped CI of the average MSE across the corpus. The CIs were calculated for all models against the Human Standard (item-wise mean of human annotations). The bootstraps create with 10,000 resamples and confidence level of 95%.

G.8.3 Bootstrapped CI of The Global Krippendorff's α

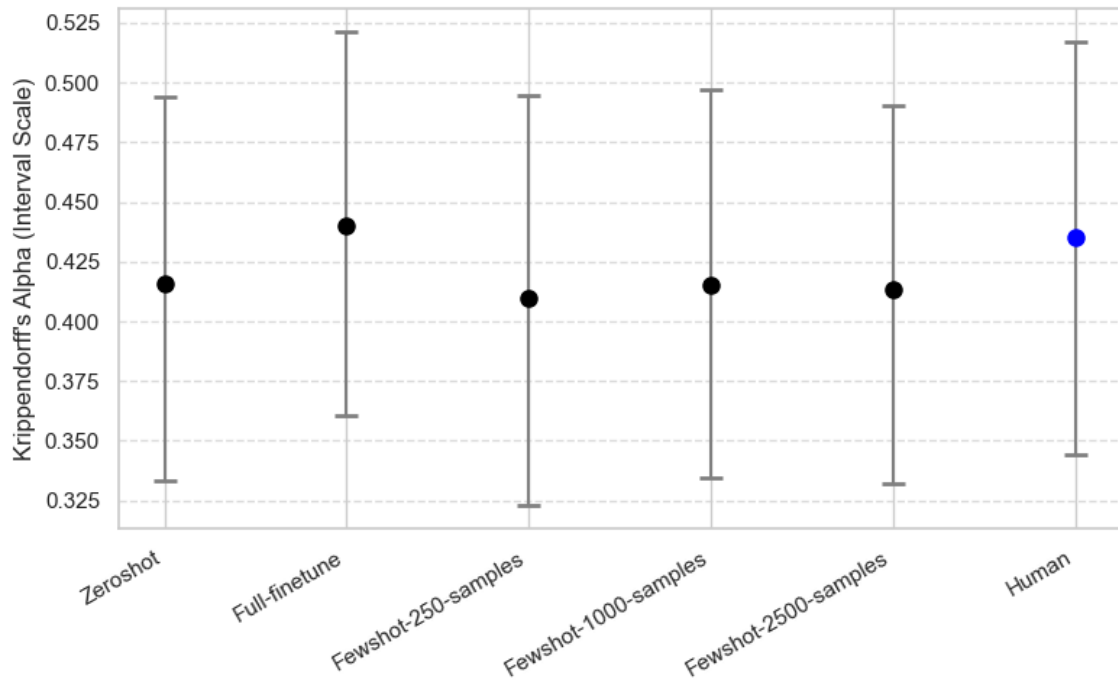


Figure G.7: Bootstrapped CI of the global Krippendorff's α . The CIs were calculated from pools including all human annotations and the rounded prediction from the marked model (item-wise mean of human annotations). The bootstraps were created with 1,000 resamples and confidence level of 95%.

H Mean scores of Authors over Samples

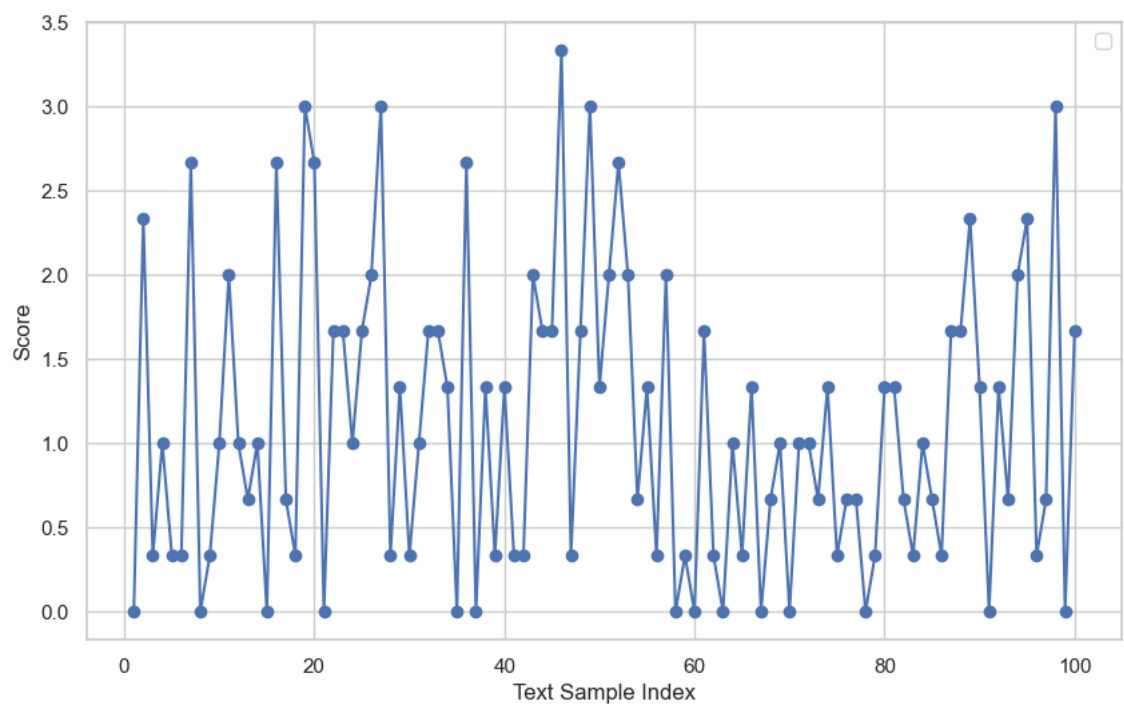


Figure H.1: Mean annotation scores of author annotators over all test text samples

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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